framework of agentic framework

7 sources

The provided documents outline a **multi-agent AI framework** designed to automate and enhance **realist systematic reviews** within complex educational systems. This system utilizes a **core-and-plugin architecture** to ensure methodological transparency, integrating **Microsoft Graph** for traceable data retrieval and **knowledge graphs** to map intricate relationships between contexts, mechanisms, and outcomes. By benchmarking the framework against the established **Richmond et al. (2020)** study, the authors demonstrate how specialized agents can replicate rigorous human synthesis while maintaining an **auditable evidence trail**. The sources emphasize a **human-in-the-loop** approach, where AI agents handle high-throughput tasks like screening and data extraction while human researchers adjudicate complex theoretical refinements. Ultimately, this research seeks to provide a scalable, **methodologically explicit infrastructure** that meets the high standards for evidence synthesis required by modern educational research.

How does pre-existing knowledge impact the success of clinical reasoning?
Explain the difference between analytical and non-analytical reasoning in students.
How can AI agents improve transparency in systematic review methods?

Today • 3:26 PM

Based on the uploaded file, it contains a multi-agent (agentic) framework that can be used to conduct a systematic literature review, consistent with the recent progress document dated February 20. I want to read through those materials to understand the core idea: how this framework can simulate the kind of research workflow that Richardman and his colleagues would carry out (the related paper should already be uploaded).

Specifically, I want the description to explain how the framework can perform the job we proposed. Before running the framework, we will conduct web crawling so the AI can access publication databases and search for relevant literature as the first step. After collecting results, we will iteratively refine and filter the retrieved literature and outputs to keep only what fits our purpose—similar to how a human researcher would screen and select papers. I will add the specific publication details later

The proposed multi-agent (agentic) framework is designed to simulate the rigorous, theory-driven research workflow of a realist review, specifically modeled after the study conducted by **Richmond et al. (2020)**. The framework's core idea is to automate the labor-intensive stages of a systematic review while preserving the

interpretive depth and methodological transparency required by human researchers.

1. Initial Harvesting: Web Crawling and Database Access

The first stage of the workflow replicates the search strategy used by Richmond and colleagues, who searched four major databases (MEDLINE, PsycINFO, ERIC, and CINAHL). In the proposed framework, a specialized **Search and Retrieval Agent** performs this job.

- **Reconstructing the Search Space:** The agent replicates the original search strings, keywords, and temporal limits to harvest candidate records from publication databases.
- **Web Crawling Integration:** This agent uses web-querying and crawling components to collect a superset of potentially relevant publications.
- **Infrastructure:** This process is powered by a **core review infrastructure** (using LangChain or LangGraph) that maintains a stable registry of all retrieved citations, ensuring every decision is traceable and auditable.

2. Iterative Refinement: Human-in-the-Loop Screening

Following the collection of results, the framework employs an **iterative filtering process** that mimics how human researchers screen papers for inclusion. This is handled by a **Screening Agent** through a two-stage process:

- **First-Stage (Title/Abstract):** The AI classifies records as *include*, *exclude*, or *uncertain* based on researcher-defined criteria encoded into its decision logic.
- **Human Calibration:** To ensure the AI accurately reflects human judgment, researchers review a small, stratified sample of the model's initial decisions. These corrections are used as **few-shot examples** to refine the agent's behavior before it re-runs the screening across the entire corpus.
- **Second-Stage (Full-Text):** For records that pass the initial screen, the pipeline attempts to access full texts and applies the same inclusion criteria using a second-stage classifier.

3. Simulating the Realist Workflow: CMOC Extraction

The most distinctive part of the Richmond workflow is the development of an **Initial Programme Theory (IPT)** and the identification of **Context–Mechanism–Outcome configurations (CMOCs)**. The agentic framework simulates this through specialized reasoning agents:

- **CMOC Extraction Agent:** This agent analyzes full-text papers to identify candidate contexts (e.g., student knowledge levels), mechanism resources (e.g., expert explanations), mechanism responses (e.g., student self-confidence), and outcomes (e.g., diagnostic accuracy).

- **Iterative Theory Refinement:** Just as Richmond et al. re-analyzed earlier studies in light of new theories, a **Synthesis and Refinement Agent** compares CMOCs across studies to identify recurrent patterns and proposes updates to the overall programme theory.
- **Knowledge Graph (KG) Support:** To better handle the complex relationships in a realist review, the framework optionally constructs a **Literature Knowledge Graph**. This allows agents to reason over connected claims and configurations—such as "why an intervention works for whom and in what circumstances"—rather than just summarizing isolated text passages.

Summary of the Framework's Job
By combining **multi-agent orchestration** with **graph-based retrieval** and **human-in-the-loop checkpoints**, the framework ensures that the AI does not merely summarize literature but actively simulates the **systematic filtering and theory-building logic** of an expert researcher. It preserves an audit trail for every action, making the results reproducible and aligned with established systematic review standards like RAMESES.
How do agents identify implicit mechanisms in the realist workflow?
Which human-in-the-loop checkpoints are used for the screening calibration?
How does GraphRAG improve retrieval compared to standard vector search?

Identify the agents: State clearly that this is a multi-agent framework and specify how many agents are included (or indicate that the number is configurable, then list the default set).

Align agents with human research roles: Explain how each agent corresponds to a specific human role in a systematic literature review and how their outputs combine to reproduce the Richman research outcomes.

Start from Richman's workflow: Before defining agents, first summarize what Richman's publication actually does: identify the tasks, the sequence of steps, and the human actions performed (e.g., search, screening, coding, synthesis, quality appraisal, reporting).

Map tasks to agents: Based on the identified human tasks, define corresponding agents and specify:

Each agent's responsibility

Inputs and outputs

Decision criteria (e.g., inclusion/exclusion rules)

How agents communicate (handoffs, shared memory, voting, arbitration)

Define agent assignments: For each agent, explicitly state what work the framework assigns to that agent (i.e., what the agent "does" in operational terms).

Prompting and control: Extend the description by explaining (at an implementation level) how the agent framework is developed using prompts:

How prompts constrain each agent's role, scope, and format

How we guide the LLM to act as the intended agent (system prompts, role prompts, tool-use prompts)

How we enforce structure (schemas, templates, rubrics) and quality control (self-check, peer review, verifier agent)

Level of detail: Provide as much operational detail as possible (agent list, workflow steps, prompt strategy, and coordination mechanism).

The proposed framework is a **multi-agent orchestration system** built using stateful workflow technology (such as LangGraph), designed to automate and scale the theory-driven processes of a realist systematic review. While the framework is **configurable** to accommodate various synthesis traditions, the default set for reproducing a realist review consists of **11 specialized agents**.

I. The Human Reference Standard: Richmond et al. (2020) Workflow

To design the framework, we first map the workflow performed by Richmond and colleagues. Their publication follows a rigorous realist synthesis process defined by the **RAMESES standards**:

1. **Theory Seeding:** Synthesizing scoping searches and expert opinion to generate an **Initial Programme Theory (IPT)**.
2. **Search Strategy:** Executing structured searches across four major databases (MEDLINE, PsycINFO, ERIC, CINAHL) using specific keywords and temporal limits (post-2000).
3. **Two-Stage Appraisal:** Screening titles and abstracts for relevance to theory building, followed by a full-text review for methodological rigour and contribution to the developing theory.
4. **Data Extraction (Coding):** Analyzing full texts to identify **Context–Mechanism–Outcome configurations (CMOCs)**.
5. **Synthesis and Refinement:** Comparing studies, identifying recurrent patterns (demi-regularities), and iteratively re-analyzing earlier studies as new evidence emerges to produce an overall programme theory.

II. Identifying the Agents and Mapping to Human Roles

The framework aligns its agents with the specific roles and cognitive tasks performed by the Richmond research team.

Agent Role	Human Research Role	Responsibility
Protocol & IPT Agent	Lead Theorist	Encodes the scope and seeding theory (IPT) to guide all subsequent agent actions.
Search & Retrieval Agent	Librarian/Searcher	Reconstructs the search space and executes queries across publication databases and enterprise folders.
Deduplication Agent	Technical Assistant	Clusters duplicate records across sources and assigns stable StudyIDs.
Title/Abstract Screener	Primary Reviewer	Applies inclusion/exclusion criteria to harvest candidate records at the metadata level.
Full-Text Acquisition Agent	Librarian	Automates retrieval of PDFs and flags missing texts for manual human upload.
Full-Text Eligibility Agent	Secondary Reviewer	Conducts second-stage screening on full texts for rigour and theory contribution.
CMOC Extraction Agent	Subject Matter Expert	Extracts contexts, mechanism resources, mechanism responses, and outcomes with evidence spans.
Theory Synthesis Agent	Synthesis Expert	Performs cross-study comparisons and proposes updates to the overall programme theory.
Contradiction Agent	Quality Auditor	Detects CMOC conflicts (e.g., same resource, different outcome) to force resolution logic.
Reporting & Artifact Agent	Technical Writer	Generates PRISMA flow counts, evidence tables, and synthesis diagrams.
Benchmark Agent	External Evaluator	Computes precision/recall by comparing results against the 28 studies in the original review.

III. Task Mapping: Inputs, Outputs, and Communication

The agents communicate through a **shared state (Study Registry, CMOC Store, and Theory Store)**, ensuring a single source of truth for the entire pipeline.

- **Communication Mechanism:** The system uses a **state-machine orchestrator** where agents hand off tasks via shared memory. For example, the *Title/Abstract Screener* writes an "include" flag to the *Study Registry*, which triggers the *Full-Text Acquisition Agent* to search for the PDF.
- **Decision Criteria:** Agents use **researcher-defined rule logic**. For the *Screening Agent*, this includes encoded criteria such as: "Population: undergraduate medical students," "Intervention: clinical reasoning training," and "Outcome: diagnostic accuracy".
- **Handoffs and Arbitration:** If the *Contradiction Agent* detects a CMOC conflict, it triggers a **Human-in-the-Loop (HITL) checkpoint** for adjudication rather than forcing a simplistic AI summary.

IV. Operational Agent Assignments
In operational terms, the framework assigns the following work:

- **Protocol Agent:** Formulates the "Initial Programme Theory" into versioned hypotheses.
- **Search Agent:** Uses **Microsoft Graph retrieval** to access institutional repositories (Teams/SharePoint) and external database connectors to harvest publications.
- **Extraction Agent:** Converts narrative passages into **structured tuples** (Context, Resource, Response, Outcome) with provenance pointers back to specific document spans.
- **Synthesis Agent:** Proposes updates to the IPT by clustering shared mechanisms or contexts across studies.

V. Prompting and Implementation Control
The framework is developed using a multi-layered prompting strategy to ensure high-quality research outputs.

1. **Role and Scope Constraints: System prompts** define the professional identity of each agent (e.g., "Act as a specialist in realist medical education synthesis"). **Role prompts** constrain the scope to specific RAMESES standards.
2. **Enforcing Structure:** The system uses **schemas and templates** to force the LLM to output structured data (like JSON or evidence tables) rather than free-form prose. **Rubrics** are embedded in the prompt to define what constitutes a valid CMOC configuration.
3. **Quality Control:**
    - **Self-Check/Reflexion:** Agents utilize a **Reflexion framework** to assess their own reasoning across multiple attempts before finalizing a decision.
    - **Verifier Agents:** A separate **Verifier Agent** audits extraction against source text to prevent hallucinations.
    - **Human Calibration:** Researchers review a stratified sample of early decisions; these corrections are recycled as **few-shot examples** to refine agent policies in real-time.

This infrastructure ensures that every action—whether the inclusion of a study or the refinement of a theory—is **auditable, traceable, and reproducible**, directly addressing the transparency gaps in traditional manual reviews.
How many agents are in the default framework?
How does the framework handle conflicting literature findings?
How are human researchers integrated into this pipeline?

Make sure to specify the agent role and specify what each agent's role will play that corresponds to the human action, particularly in that Richmond paper and how Richmond and his colleagues and team did to conclude their systematic review. And also, you need to clearly identify which is the core part of the framework and which is the foundation of understanding effective relations and find the ground truth from the publication that is the core framework. And then the plug-in framework that is aligned with the specific review method, logistics, logic, and review logic. And make it clear how they are distinguishing them and how they can be realized using multiple agents in a large language model framework. What is the template? What are the prompts and the template? Give some examples of prompts for functions, particularly agents. What is the input of the agent? What is the output of the agent? How does the human verify the output and provide feedback to the system, so that the system can modify the outcome based on the human's comments and evaluation.

The proposed multi-agent (agentic) framework is a stateful orchestration system designed to automate the theory-driven processes of a systematic review while maintaining the methodological rigor of a human research team. To understand its core idea, we must first map it against the research workflow established by **Richmond et al. (2020)**.

I. The Reference Standard: Richmond et al.'s Workflow

The study by Richmond and colleagues followed the **RAMESES standards** for realist synthesis to understand how educational interventions develop clinical reasoning. Their human-led workflow involved the following actions:

1. **Theory Seeding:** The lead author (AR) synthesized scoping searches and expert consensus from the team (RP, SG, NC) to generate an **Initial Programme Theory (IPT)**.
2. **Structured Searching:** They searched four major databases (MEDLINE, PsycINFO, ERIC, CINAHL) for studies published from 2000 onwards.
3. **Two-Stage Screening:** A title-abstract screen for relevance to theory building, followed by a full-text review to assess methodological rigour and contribution to the theory.
4. **Data Extraction (Coding):** They analyzed **28 included papers** to identify **Context–Mechanism–Outcome configurations (CMOCs)**.
5. **Synthesis and Refinement:** Studies were compared, recurrent patterns (demi-regularities) were identified, and earlier studies were re-analyzed in light of new theories to refine the overall programme theory.

II. Core Framework vs. Plugin Framework

The framework distinguishes between the "infrastructure" of a review and its "intellectual logic":

- **The Core Infrastructure (Method-Agnostic):** This is the foundation of the framework. It handles the "plumbing" common to nearly all systematic reviews: corpus ingestion, Microsoft Graph-enabled retrieval, study registry management, deduplication, and the **Human-in-the-Loop (HITL) control layer**.
- **The Plugin Framework (Method-Specific):** This is the **Realist Synthesis Plugin**, which aligns with Richmond's specific review logic. It contains the rules for IPT management and CMOC extraction.
- **Foundation of Effective Relations (The Knowledge Graph):** To find the "ground truth" and understand complex relations (why/for whom/in what circumstances), the framework constructs a **Literature Knowledge Graph (LKG)**. While the LLM provides interpretive reasoning, the LKG supplies the structured, graph-grounded evidence of causal configurations (CMOCs) that are often lost in flat text summaries.

III. Identifying the Agents and Roles

The framework assigns work to specialized agents that correspond to Richmond's team roles:

Agent Role	Human Action Corresponded	Operational Assignment
Protocol & IPT Agent	Lead Theorist (AR)	Encodes the scope, IPT hypotheses, and extraction schemas.
Search & Retrieval Agent	Librarian/Searcher	Executes queries via Microsoft Graph API across databases and Teams/SharePoint folders.
Deduplication Agent	Technical Assistant	Clusters duplicate records and assigns stable StudyIDs.
Title/Abstract Screener	Primary Reviewer	Classifies records as *include, exclude,* or *uncertain* based on eligibility rules.
Full-Text Eligibility Agent	Secondary Reviewer (RP/SG/NC)	Applies inclusion criteria to retrieved PDFs for rigour and theory contribution.
CMOC Extraction Agent	Subject Matter Expert	Extracts structured tuples: **Context** (student/teacher level), **Mechanism** (resource/response), and **Outcome**.
Synthesis & Refinement Agent	Synthesis Expert	Compares CMOCs across studies to identify demi-regularities and update the programme theory.
Contradiction Agent	Quality Auditor	Detects CMOC conflicts (e.g., same resource leading to different outcomes) to force resolution.
Benchmark Agent	External Evaluator	Computes recall/precision by comparing model results to Richmond's 28 studies.

IV. Prompts, Templates, and Implementation

The framework is realized using **Large Language Models (LLMs)** orchestrated through a state-machine (like LangGraph).

The Prompting Strategy:

- **System Prompts:** Constrain the agent's role and scope (e.g., "Act as a specialist in RAMESES-compliant realist synthesis").
- **Structured Templates/Schemas:** Ensure outputs are structured (JSON/tables) rather than narrative, making them auditable.

Agent Examples:

- **Function:** CMOC Extraction
- **Input:** Full-text PDF chunk + IPT Schema.
- **Prompt Example:** *"Identify the student-level Context (C), the Mechanism Resource provided (M-res), the student's cognitive Response (M-resp), and the Reasoning Outcome (O). Link each to a specific text span."*.
- **Output:** A structured JSON object containing the CMOC configuration with provenance pointers.

V. Human Verification and Feedback Loop
The framework is designed for **lifelong (continual) learning** through human interaction.

1. **Checkpoints:** Humans intervene at predefined stages: **screening adjudication**, **CMOC validation**, and **theory sign-off**.
2. **Calibration:** In the screening stage, the human reviews a small sample of the agent's decisions. Corrections are recycled into the system as **few-shot examples** to refine the agent's logic for the remaining corpus.
3. **Feedback & Modification:** If a researcher is unsatisfied with a synthesis outcome (e.g., an extraction is too generic), they provide feedback. The system then updates its **decision policies** (prompts or rule logic) and reruns the relevant agent to modify the outcome.

This interactive process ensures the system accommodates human requirements and adapts as the researcher's understanding of the theory evolves.
How do agents handle contradictions in realist synthesis?
Tell me more about the human-in-the-loop calibration step.
What analytical artifacts does the framework export for repositories?

Okay, I said that the framework should not miss the interaction with the human at each step, and how the human decision rules will play a role in this framework. Also, you should not include a final agent that compares the machine's outcome with the human's outcome. In a real, practical framework, this step does not exist. We only do model-vs-human comparison manually or separately (independent of the model) for evaluation/research purposes to verify the model — not as part of the model in practical application. For the model in practical application, there should be a channel

or points where people can review and audit the outcomes and procedures, provide feedback, and then have the system redo or improve the results based on that feedback before moving forward. Yeah, that is what we want.

In the practical application of this multi-agent framework, the system is designed as a **Human-in-the-Loop (HITL) orchestration pipeline**, where human researchers are not passive observers but active governors of the review's logic and outcomes. The framework is partitioned into a **method-agnostic core review infrastructure** and a **method-specific synthesis plugin** (e.g., the Realist Synthesis plugin), ensuring that every automated step is governed by human-defined decision rules.

I. Operational Multi-Agent Framework (Practical Model)

The following agents constitute the operational set. Per your requirement, the "Benchmark Agent" is excluded as it serves purely for separate evaluation and research verification rather than live production.

1. **Protocol & IPT Agent:** Encodes the researcher's initial hypotheses and inclusion rules into versioned templates.
2. **Search & Retrieval Agent:** Uses **Microsoft Graph** to pull search strings and literature from enterprise folders (SharePoint/Teams) while logging every query for auditability.
3. **Deduplication Agent:** Clusters duplicate records and ensures a stable study registry.
4. **Title/Abstract Screening Agent:** Classifies records based on human-defined eligibility criteria.
5. **Full-Text Eligibility Agent:** Re-evaluates studies at the full-text level for methodological rigour and theory contribution.
6. **CMOC Extraction Agent:** Deconstructs text into Context, Mechanism, and Outcome configurations.
7. **Synthesis & Refinement Agent:** Proposes updates to the programme theory by comparing CMOCs across the corpus.
8. **Contradiction & Sensitivity Agent:** Surfaces conflicting evidence (e.g., same intervention leading to different outcomes) for human adjudication.
9. **Reporting & Artifact Agent:** Generates audit-ready evidence tables, PRISMA counts, and theory diagrams.

II. Human Interaction Points and Feedback Loop

The framework replaces the final "automated comparison" with **predefined interaction checkpoints** where researchers audit outcomes, provide feedback, and trigger the system to redo or improve results before proceeding.

- **Checkpoint 1: Screening Calibration & Adjudication**

    - **Action:** Researchers review a stratified sample of the AI's screening decisions.
    - **Feedback/Redo:** If the AI misinterprets a criterion (e.g., "clinical reasoning" vs. "general knowledge"), the researcher provides corrections. These corrections are recycled as **few-shot examples** to refine the agent's decision logic, and the screening agent **reruns** the task on the affected corpus to improve accuracy.
- **Checkpoint 2: CMOC Extraction Validation**
    - **Action:** Humans spot-check the extracted CMOC tuples to ensure they reflect the **"ground truth"** found in the publication.
    - **Feedback/Redo:** Researchers can modify the extraction schema or specific prompt templates if the AI is missing implicit mechanisms. The system then **re-extracts** data using the updated rules.
- **Checkpoint 3: Contradiction Adjudication**
    - **Action:** When the system identifies conflicting evidence (e.g., an intervention works for experts but fails for novices), it stops and presents a **contradiction register** to the researcher.
    - **Feedback/Redo:** The human provides an interpretive explanation (e.g., "this is a context-specific moderating factor"). This feedback is stored in the **Theory Store**, and the synthesis agent incorporates this logic into the final programme theory.
- **Checkpoint 4: Programme Theory Sign-off**
    - **Action:** Before generating final artifacts, the human reviews the entire evidence path from theory claim to specific text spans.
    - **Feedback/Redo:** The researcher can reject theory updates that lack interpretive depth, forcing the Synthesis Agent to re-analyze the **Literature Knowledge Graph** using a different reasoning structure.

III. Lifelong (Continual) Learning Mechanism

This interaction protocol enables **lifelong (continual) learning** within the model. By capturing human feedback at each checkpoint, the framework iteratively updates its **decision policies** (prompts, rubrics, and rule-logic). This ensures the framework accommodates evolving stakeholder requirements and adapts its behavior throughout the lifecycle of the review.

IV. Transparency and Auditability

Every human intervention, feedback note, and system "redo" is logged in the **Audit & Repository Export module**. This creates a complete, transparent record of how human expertise and AI automation combined to produce the synthesis, fulfilling the RRE Call's requirement for **methodologically explicit guidelines** and verifiable analytical artifacts.

How does the realist synthesis plugin ensure RAMESES standard compliance?

What specific audit artifacts are exported for the RRE repository?
How do the human-in-the-loop checkpoints improve CMOC extraction quality?

combine the above agentic framework with the above modificiation to layout the framework, with detailed how each agent will be prompted and provide prompt example or template

The proposed multi-agent framework is a stateful orchestration system designed to automate the labor-intensive stages of a systematic review while maintaining the interpretive depth of the **Richmond et al. (2020)** study. The system consists of **10 specialized agents** in its default configuration, which are coordinated through a **Core Review Infrastructure** and a method-specific **Realist Synthesis Plugin**.

I. The Richmond Human Reference Standard

Before mapping the agents, it is necessary to identify the specific tasks performed by Richmond and colleagues:

- **Theory Seeding:** The team synthesized expert opinion to generate an **Initial Programme Theory (IPT)**.
- **Structured Search:** They conducted keyword searches across four major databases (MEDLINE, PsycINFO, ERIC, CINAHL) for studies published from 2000 onwards.
- **Two-Stage Screening:** Reviewers performed a title/abstract screen for relevance, followed by a full-text review for methodological rigour.
- **CMOC Extraction:** The team analyzed 28 papers to extract **Context–Mechanism–Outcome configurations (CMOCs)**.
- **Iterative Synthesis:** They compared CMOCs across studies to identify "demi-regularities" and refined the IPT into a final programme theory.

II. Core Infrastructure vs. Method Plugin

The framework distinguishes between universal review "plumbing" and intellectual synthesis logic:

- **Core Review Infrastructure:** Handles method-agnostic tasks such as **Microsoft Graph-enabled retrieval**, deduplication, study registry management, and the **Human-in-the-loop (HITL) control layer**.
- **Realist Synthesis Plugin:** Provides the method-specific "semantics," including rules for **CMOC extraction** and **theory refinement**.

III. Detailed Agent Layout and Prompting Strategy

The agents communicate through a **shared state** (Study Registry and Literature Knowledge Graph) where one agent's output triggers the next.

Agent Role	Richmond	Input	Output	Operational

	Alignment			Assignment
Protocol & IPT Agent	Lead Theorist (AR)	Review Question/Scope	IPT Schema & JSON Template	Encodes IPT hypotheses to guide extraction rules.
Search & Retrieval Agent	Librarian/Searcher	IPT Themes/Keywords	Raw Publication Superset	Executes Microsoft Graph-based queries across indexed literature.
Deduplication Agent	Tech Assistant	Raw Superset	Unique Study Registry	Clusters duplicates and assigns stable StudyIDs.
Title/Abstract Screener	Primary Reviewer	Study Metadata	Decision (Inc/Exc/Unc)	Classifies records based on eligibility rules; triggers HITL if "Uncertain".
Full-Text Acquisition Agent	Librarian	"Include" Decisions	PDF/Full Text Chunks	Automates retrieval and flags missing PDFs for human upload.
Full-Text Eligibility Agent	Secondary Reviewer	Full Text PDFs	Eligibility Report	Applies rigour/theory contribution criteria at the full-text level.
CMOC Extraction Agent	Subject Expert	Full Text + IPT Schema	LKG Triples (C-M-O)	Extracts structured configurations with evidence-span links.
Theory Synthesis Agent	Synthesis Expert	LKG Subgraphs	Refined Programme Theory	Clusters CMOCs into recurrent patterns to update theory.
Contradiction Agent	Quality Auditor	CMOC Store	Conflict Register	Detects CMOC conflicts (e.g., same resource, different outcome) for human adjudication.
Reporting Agent	Tech Writer	Shared State	PRISMA Flow/Evidence Tables	Generates auditable artifacts for repository deposit.

IV. Prompting and Control Implementation

The framework uses **Role Prompts** to define identity and **Schemas** to enforce structure.

1. Role and Context Constraints:

- **System Prompt Template:** *"Act as a specialist in RAMESES-compliant realist synthesis. Your goal is to identify how educational interventions develop*

clinical reasoning. You must follow researcher-defined inclusion rules strictly and link every claim to a specific document section ID."

2. Enforcement of Structure (CMOC Extraction Example):

- **Input Agent Prompt:** *"Analyze the following text chunk. Extract CMOC configurations according to this schema: {Context: Student Knowledge Level, Mechanism_Resource: Pedagogical Tool, Mechanism_Response: Cognitive Reaction, Outcome: Diagnostic Accuracy}. If a mechanism is implicit, state your reasoning for the inference."*
- **Output Format:** JSON-structured tuples mapped to the **Literature Knowledge Graph (LKG)**.

3. Quality Control (HITL and Redo Triggers):

- **Calibration Checkpoint:** For the Screening Agent, the human reviews a 5% stratified sample. **Feedback Prompt:** *"The human researcher marked StudyID 123 as 'Exclude' because it was postgraduate, not undergraduate. Update your decision logic and re-screen the remaining records."*
- **Adjudication Checkpoint:** When the **Contradiction Agent** flags a conflict, it stops the pipeline. The human provides a resolution (e.g., *"This outcome is moderated by the teacher's experience"*), which is added to the **Theory Store** as a new rule for the Synthesis Agent to "redo" its theory refinement.

This interaction ensures that the system accommodates evolving human requirements and creates a **traceable evidence path** from theory claims back to verifiable source passages, fulfilling the transparency requirements of the RRE call.
How will the framework handle implicit mechanisms during extraction?
Can you elaborate on the repository artifacts required by AERA?
What criteria will experts use to evaluate CMOC validity?

Please lay out the details of the framework so that people can clearly see:

How each agent will be prompted What is the input for each large language model agent What is the expected output of each agent How the output of one agent is passed / loaded / sent to the next agent in the workflow

Among all the agents, please clarify:

Which agent(s) are the core of the framework (essential / always present) Which agent(s) belong to the plug-in functions (optional / modular / aligned with specific review methods)

Thank you.

The details of the multi-agent framework are structured to separate the **Core Review Infrastructure** (essential plumbing) from **Method-Specific Plugins** (specialized synthesis logic). This architecture ensures that the system can be adapted to various review traditions while maintaining a stable, auditable foundation.

I. Core Review Infrastructure (Essential / Always Present)

These agents provide method-agnostic services required for any systematic review. They are governed by a **stateful orchestrator** (e.g., LangGraph) that manages a **Shared Study Registry**—the "single source of truth" for the pipeline.

Agent Role	Input	Expected Output	Prompting Strategy
Search & Retrieval Agent	Search strings and IPT themes.	Raw publication superset with metadata.	**Role Prompt:** "Act as a librarian executing Microsoft Graph queries across Teams, SharePoint, and external databases".
Deduplication Agent	Raw publication list.	Unique Study Registry with stable **StudyIDs**.	**Template Prompt:** "Compare record A and B; if identical, merge metadata and assign a primary StudyID".
Title/Abstract Screener	Study metadata (Title/Abstract) + Eligibility rules.	Label (Include/Exclude/Uncertain) + Short Rationale.	**Schema-Enforced Prompt:** "Evaluate against the criteria: [Criteria]. Output JSON with rationale and confidence score".
Full-Text Acquisition	"Include" decisions and StudyIDs.	Full-text PDFs or text chunks; flags missing files.	**Tool-Use Prompt:** "Search M365 folders via Microsoft Graph for the PDF matching StudyID X".
Full-Text Eligibility	Full-text documents + Eligibility rules.	Final eligibility report; justifies exclusions.	**Role Prompt:** "Secondary Reviewer: Audit the full text for methodological rigour and theoretical fit".
Reporting & Audit Agent	Full Study Registry and shared state.	PRISMA flow counts, evidence tables, and artifact logs.	**Formatting Prompt:** "Generate a RAMESES-compliant audit trail from the decision logs for repository deposit".

II. Method-Specific Plugins (Realist Synthesis Plugin)

These agents are modular and aligned with the specific logic of the **Richmond et al. (2020)** benchmark. They reason over a **Literature Knowledge Graph (LKG)** to find configurations of causal mechanisms.

Agent Role	Input	Expected Output	Prompting Strategy

Protocol & IPT Agent	Review questions and seeding expertise.	Initial Programme Theory (IPT) hypotheses and extraction schemas.	**Theory Seeding Prompt:** "Synthesize the provided expertise to generate an IPT focused on student-level clinical reasoning".
CMOC Extraction Agent	Full-text chunks + IPT Schema.	Structured **Context–Mechanism–Outcome** tuples with evidence spans.	**Inference Prompt:** "Identify implicit responses (Mechanism-Response) from the text. Link each claim to a specific span ID".
Synthesis & Refinement	Extracted CMOCs from the LKG.	Refined Programme Theory (Updated IPT) and demi-regularity patterns.	**Pattern Recognition Prompt:** "Compare CMOCs across studies. Propose an update to the IPT if context X consistently alters outcome Y".
Contradiction Agent	LKG subgraphs.	**Contradiction Register** identifying conflicting evidence.	**Conflict Detection Prompt:** "Flag instances where the same resource leads to different outcomes. Present for human adjudication".

III. Workflow Orchestration and Handoffs

The framework does not use simple linear passing; instead, it uses **Stateful Handoffs** via the **Shared Registry** and **Literature Knowledge Graph**.

- **Passing Mechanisms:** When the *Deduplication Agent* writes a unique StudyID to the registry, the *Screening Agent* is automatically triggered to process that record.
- **Loading Evidence:** The *Extraction Agent* loads full-text chunks and pushes extracted "triples" (entities and relations) into the **Literature Knowledge Graph (LKG)**.
- **Sending for Feedback:** If the *Contradiction Agent* detects a conflict, it pauses the synthesis and sends a notification to the **Human-in-the-Loop (HITL) control module**.

IV. Human-in-the-Loop and Lifelong Learning

The human researcher interacts with the framework at four predefined **checkpoints**:

1. **Screening Adjudication:** Human reviews a stratified sample of AI decisions and provides corrections.

2. **Extraction Validation:** Human spot-checks extracted CMOCs for "ground truth" fidelity.
3. **Contradiction Resolution:** Human provides the interpretive logic to explain conflicting findings.
4. **Theory Sign-off:** Human approves the final synthesized programme theory before artifact generation.

Lifelong Update Mechanism: Human feedback at these points is recycled to update the system's **decision policies** (prompts and rules). This allows the agents to "learn" and accommodate evolving stakeholder requirements as the review moves into new cycles or as new studies appear.
How does the framework automate realist CMOC extraction?
Explain the role of human-in-the-loop checkpoints in the pipeline.
How do knowledge graphs improve reasoning over flat text summaries?

Please look at the framework. We particularly emphasize that human decisions will dominate this process. There are multiple checkpoints to let humans review the process, provide feedback, and ensure the system follows the human's decisions with human oversight and master control over this process. Explain how this is integrated into those frameworks and how the machine will incorporate the human response to revise its efforts through an iterative process. Please give a good picture, not only based on the content provided, but also based on your knowledge and an extensive search of the internet to propose something that goes beyond the provided material.

In this multi-agent framework, human researchers do not simply provide an initial input and wait for a result; they act as the **master controllers** of an iterative, state-aware system. The agents are designed as specialized assistants that handle the labor-intensive "plumbing" of a review while pausing at critical junctures for **human adjudication and policy refinement**. This integration is realized through a series of "Master Control" checkpoints and a feedback-to-policy loop that enables the system to learn from human corrections in real-time.
1. Integration of Human Oversight: The Four Checkpoints
The framework incorporates human judgment at four predefined "gates" where the automated process halts until a human provides a "sign-off" or corrective feedback.

- **Checkpoint 1: Screening Calibration (The "Few-Shot" Loop):** Instead of screening the entire corpus blindly, the **Screening Agent** first classifies a small, stratified sample of records. The human reviews these decisions, correcting any misclassifications. These corrections are then automatically converted into **few-shot examples** that the system uses to refine its internal decision logic before re-running the screening across the remaining thousands of papers.

- **Checkpoint 2: CMOC Validation (The "Ground Truth" Audit):** For a realist review like the **Richmond et al. (2020)** study, identifying Context–Mechanism–Outcome configurations (CMOCs) is an interpretive act. Humans spot-check extracted CMOC tuples to ensure they reflect the **"ground truth"** of the publications. If a human identifies that an agent is missing implicit mechanisms, they can update the extraction schema or the specific **role prompt** for that agent.
- **Checkpoint 3: Contradiction Adjudication (The "Conflict Registry"):** When the **Contradiction Agent** identifies conflicting evidence—such as the same educational intervention leading to different outcomes in different settings—it pauses the pipeline and presents a "conflict register" to the researcher. The human provides an interpretive explanation for the conflict (e.g., explaining why "low knowledge" students respond differently than experts), which the system then incorporates into the overall **Programme Theory**.
- **Checkpoint 4: Theory Sign-off:** Before final reporting, the researcher reviews the entire evidence path. They can reject updates to the theory that lack interpretive depth, forcing the **Synthesis Agent** to "redo" its comparative analysis based on new constraints.

2. The Iterative Revision Process: Feedback-to-Policy
The machine incorporates human response not just as a one-time correction, but as a **revision of its future efforts** through two primary mechanisms:

- **Dynamic Prompt & Rule Revision:** Human feedback is used to iteratively update the system's **decision policies** (e.g., prompts, rubrics, and extraction rules). For example, if a researcher provides feedback that "clinical reasoning" should include "diagnostic accuracy," the **Protocol Agent** updates the project-wide rules, and all downstream agents (Search, Screen, and Extract) adapt their behavior to this new master definition.
- **The "Redo" Trigger:** The framework utilizes a **state-machine orchestrator** (such as LangGraph) that maintains a "Shared Registry". If a human provides negative feedback at any checkpoint, the orchestrator can trigger a **recursive "redo" loop**. This means the system doesn't just edit the text; it re-invokes the relevant agent with the updated human instruction, ensuring the entire evidence chain remains consistent and auditable.

3. Beyond the Provided Material: Proposing "Master Control" Innovations
To move beyond current implementations, the framework can integrate the following "Master Control" concepts:

- **Semantic Policy Anchoring:** Instead of just using "fixed prompts," the system could maintain a **"Living Project Handbook"**—a centralized knowledge store of all human decisions and comments. When an agent encounters an ambiguous study, it would first query this handbook to see how the human handled similar cases previously, ensuring that the machine's "internal policy" remains perfectly aligned with the human's evolving intent.
- **Active Preference Learning:** The system could utilize **Reinforcement Learning from Human Feedback (RLHF)** specifically tuned for evidence synthesis. By observing which studies a researcher prioritizes and which CMOCs they "upvote" during validation, the agents could autonomously adjust their retrieval and synthesis strategies to match the researcher's specific **intellectual style and theoretical focus**.
- **Master Intervention UI:** A specialized dashboard—an **Adjudication UI**—would allow researchers to see the "uncertainty score" for every AI decision. By sorting the workload by uncertainty, the human can focus their limited time on the most contested pieces of evidence, providing the high-level reasoning that the machine lacks, while the AI handles the bulk of the repetitive tasks.

Summary of Human Master Control
By integrating **predefined checkpoints**, **stateful redo-loops**, and **lifelong learning mechanisms**, the framework ensures that human decisions dominate the process. The machine acts as an **auditable executor** of human intent, where every revision is logged, traceable to a specific human comment, and verified against the original publication evidence.
How do these plugins specifically manage different synthesis logics?
Could you explain the specific artifacts produced for auditability?
How does human feedback improve the system's decision policies?

How do these plugins specifically manage different synthesis logics?

In the proposed multi-agent framework, the management of diverse synthesis logics is achieved through a modular **"core-and-plugin" architecture**. While the **core review infrastructure** handles method-agnostic tasks like retrieval and deduplication, the **synthesis plugins** instantiate the specific intellectual logic, rules, and "synthesis semantics" of a particular review tradition.
These plugins specifically manage different synthesis logics through three primary mechanisms:
1. Method-Specific Schemas and Extraction Rules
Each plugin provides the **schema definitions** and **rule-based decision logic** that constrain how agents extract and interpret evidence. For example:

- **Realist Synthesis Plugin:** Operationalizes **Initial Programme Theory (IPT)** and **Context–Mechanism–Outcome Configurations (CMOCs)**. Agents are prompted specifically to identify "mechanism resources" and "mechanism responses," even when they are implicit in the text.
- **Thematic/Framework Synthesis Plugin:** Directs agents to code findings into pre-specified frameworks or translate study concepts into emergent themes.
- **QCA (Qualitative Comparative Analysis) Plugin:** Focuses on **case calibration** and the discovery of necessity/sufficiency patterns among various configurations of causal factors.

2. Relational Reasoning via Literature Knowledge Graphs (LKG)
Plugins utilize a **Literature Knowledge Graph** to manage the complex, non-linear relationships required by advanced synthesis methods. While a standard search might only retrieve text chunks, the plugin enables **Graph-enhanced Retrieval-Augmented Generation (GraphRAG)**.

- In a **Realist logic**, this allows agents to query the graph for "subgraphs" that link specific context factors to mechanism responses.
- For **complex-system synthesis**, this enables the framework to surface "configurations" and "contingencies" rather than just isolated findings.

3. Iterative Refinement and Contradiction Handling
The plugins manage the **iterative synthesis loops** unique to their specific methodology.

- **Theory Refinement:** In a realist review, the plugin manages a "Theory Store" that versions the programme theory. As new CMOCs are identified, a specialized **Synthesis Agent** compares them against the current theory and proposes refinements.
- **Contradiction Adjudication:** The **Contradiction Agent** identifies conflicts in the evidence—such as the same intervention leading to different outcomes in different contexts—and presents these to a human for adjudication. The plugin then incorporates this human explanation back into the synthesis logic as a "contextual contingency".

4. Human-in-the-Loop (HITL) Policy Updates
Plugins define the specific **interaction protocol** where human researchers oversee the synthesis logic. Through **lifelong (continual) learning**, human feedback at method-specific checkpoints (like CMOC validation) is used to update the plugin's **decision policies, prompts, and templates**. This ensures the framework's "synthesis logic" evolves to match the researcher's intent as the review progresses.
How do multi-agent frameworks handle complex causal configurations?

What repository artifacts are required for RRE compliance?
How does human feedback refine the AI's screening logic?

Sunday, February 22

Please address each of the specific plug-in methods for the systematic review based on the uploaded call for papers. Include all drafts and proposals regarding the different specific review requirements, and give more details on each method: how the plug-in will align with that specific systematic review, how the methods differ from each other, and the different perspectives of systematic reviews. Can you find the proposals and those relevant contents in the uploaded proposal or uploaded call for papers issued by the Journal Editor?

The proposed multi-agent framework is designed around a **core-and-plugin architecture** specifically to meet the diverse methodological requirements of the **Review of Research in Education (RRE) 2026 call**. While the core infrastructure handles method-agnostic tasks like provenance-tracked retrieval and study registry, the **method-specific synthesis plugins** instantiate the distinct review logics needed to address complex educational systems.

1. Realist Synthesis Plugin (Primary Worked Example)

As established in the source materials, Realist Synthesis serves as the primary anchor for the framework because its focus on **Context–Mechanism–Outcome configurations (CMOCs)** directly aligns with the RRE call's emphasis on "interacting elements," "causal mechanisms," and "contextual contingencies".

- **Alignment:** This plugin operationalizes the **Initial Programme Theory (IPT)** and iterative theory refinement, addressing the "why, how, and for whom" questions essential for informing change in complex systems.
- **Agentic Logic:** Specialized reasoning agents extract structured configurations (CMOCs) from literature, linking mechanism resources (what an intervention provides) to learner responses and subsequent outcomes.
- **Core Difference:** Unlike descriptive methods, this plugin is **theory-driven and explanatory**, focusing on the generative causation that occurs when interventions interact with specific contexts.

2. Qualitative Comparative Analysis (QCA) Plugin

The RRE call highlights the need for methods that can reveal "configurations of causal mechanisms".

- **Alignment:** A QCA plugin supports **causal discovery** by identifying which combinations of conditions are necessary or sufficient for an outcome.

- **Agentic Logic:** This plugin manages **case calibration**—transforming qualitative or quantitative data into set relations—and applies set-theoretic logic to find patterns that simple aggregation might miss.
- **Core Difference:** It differs from realist synthesis by focusing on **set-theoretic configurations** rather than the narrative "mechanism-response" logic.

3. Framework and Thematic Synthesis Plugins
These plugins are designed for the structured coding of large evidence bases to identify recurring patterns and demi-regularities.

- **Alignment:** These methods provide the "systematic approach" required by the call to ensure that conclusions are warranted through transparent and auditable coding.
- **Agentic Logic:**
  - **Framework Synthesis:** Agents use a **pre-specified framework** to code evidence, ensuring alignment with existing educational theories.
  - **Thematic Synthesis:** Agents identify **emergent themes** through inductive coding, allowing for the synthesis of heterogeneous studies.
- **Perspective:** These represent a **pattern-seeking perspective**, where the goal is to aggregate findings across diverse study designs into a coherent set of themes or categories.

4. Meta-Ethnography and Critical Interpretive Synthesis (CIS) Plugins
The call invites "diverse approaches to research," including those emphasizing theoretical critique.

- **Alignment:** These plugins address the "theorize and warrant conclusions" requirement of the call by moving beyond mere summary toward **theoretical innovation**.
- **Agentic Logic:** Agents perform **conceptual translation** across studies, identifying how concepts from one study can be reconciled or expanded by another to build a new overarching theory.
- **Perspective:** These represent a **highly interpretive perspective**, prioritizing theoretical depth and critique over the procedural standardization of meta-analysis.

5. Meta-Analysis and Mixed-Method Synthesis Plugins
To satisfy the call's emphasis on "methodologically diverse evidence," the framework supports quantitative and integrative plugins.

- **Alignment:** These plugins provide the rigor needed to handle the "measurable effects" of interventions while integrating them with qualitative insights to inform systemic change.
- **Agentic Logic:**
    - **Meta-Analysis:** Specialized agents extract **effect sizes**, perform risk-of-bias assessments, and apply quantitative pooling models.
    - **Mixed-Method:** The framework embeds text and numeric data into **unified graph structures** (Literature Knowledge Graphs), allowing agents to reason across diverse data types for causal discovery.
- **Difference:** These methods differ by their reliance on **statistical pooling** (meta-analysis) or **integrative triangulation** (mixed-methods), providing a quantitative baseline that interpretive plugins lack.

Summary of Plugin Alignment with RRE Requirements
The sources emphasize that by maintaining a **method-neutral core**—which provides the audit trails and provenance required by RRE standards—the framework can remain a "living" methodological infrastructure. Each plugin allows the system to switch its **reasoning substrate**: for a realist synthesis, it queries the knowledge graph for CMOC subgraphs; for QCA, it queries for set-theoretic configurations; and for thematic synthesis, it clusters emergent semantic embeddings. This ensures that the AI is not merely summarizing literature but is performing the specific **analytical and warranting tasks** defined by each unique research tradition.
How do core infrastructure and synthesis plugins work together?
What specifically are the methodologically explicit guidelines for these plugins?
How does the framework manage contradiction adjudication in complex systems?

could you expand the above? Could you make details for each plugin as a methodology method? This is a method in the agent-based model agent framework. Please, for each one, provide a detailed method. And then, in this paper, we only work on one method, but we need to lay out the method or method framework for the other plugins or other systematic reviews. Do you understand? So look at the call for papers and the uploaded proposal that is uploaded for this draft paper, the proposal for the draft paper, and lay out this plugin as part of the methodology. That is the core part of this draft paper. And also, we need to make sure we have the core, C-O-R-E, core part of that framework.

The following methodology section details the architecture of the proposed multi-agent system, distinguishing between the method-agnostic **Core Review Infrastructure** and the method-specific **Synthesis Plugins**. This architecture is designed to satisfy the **Review of Research in Education (RRE) 2026** requirements for methodological innovation and transparent warranting in complex educational systems.

4. Methodology: The Multi-Agent Framework

This method employs a stateful orchestration architecture (implemented via LangGraph) where a specialized set of agents collaborates under a **Human-in-the-Loop (HITL)** governance model. The framework is partitioned into two distinct layers: a method-agnostic core and a method-specific plugin layer.

4.1 The Core Review Infrastructure (CORE)

The **CORE** provides the foundational services required for any systematic review, ensuring that every automated step is auditable, reproducible, and grounded in institutional data. It consists of the following essential modules:

- **Protocol & Scope Module:** Encodes researcher-defined review questions, inclusion/exclusion rules, and extraction schemas into versioned prompt templates.
- **Retrieval & Provenance Module:** Utilizing **Microsoft Graph** as the enterprise retrieval layer, this module executes search strings and harvests literature from project folders (Teams/SharePoint) while logging every query and source for absolute auditability.
- **Study Registry & Deduplication Module:** Clusters harvested records, assigns stable **StudyIDs**, and maintains a "Single Source of Truth" registry for the entire pipeline.
- **Screening Workflow Module:** A multi-stage classifier applying eligibility rules at title/abstract and full-text levels. It categorizes studies as *include, exclude,* or *uncertain* while generating detailed rationales for every decision.
- **Extraction & Evidence-Linking Module:** Deconstructs documents into structured data objects. Every extracted item is persistently linked (provenance) to a specific text span ID to prevent hallucinations and support manual verification.
- **Audit & Repository Export Module:** Automatically packages all decision logs, search queries, extraction tables, and human feedback records into analytical artifacts ready for deposition in the AERA-RRE repository.

4.2 Synthesis Plugin Framework

Synthesis plugins are modular extensions that instantiate the intellectual logic and coding rules of a specific review tradition. While this manuscript focuses on the **Realist Synthesis plugin** as the primary worked example, the framework is designed to support the diverse methodological landscape required by the RRE call.

4.2.1 Realist Synthesis Plugin (Worked Example)

This plugin operationalizes the **RAMESES** standards to explain *why, how, and for whom* interventions develop clinical reasoning ability.

- **Methodology:** It manages the **Initial Programme Theory (IPT)** and drives agents to extract **Context–Mechanism–Outcome configurations (CMOCs)**.

- **Extraction Logic:** Specialized agents extract structured tuples: **Context** (e.g., learner knowledge level), **Mechanism Resource** (the pedagogical tool provided), **Mechanism Response** (the cognitive/emotional reaction), and the resulting **Outcome** (e.g., diagnostic accuracy).
- **Iterative Refinement:** Agents query a **Literature Knowledge Graph** to identify recurrent patterns (demi-regularities) across studies and propose evidence-based updates to the overall programme theory.

4.2.2 Framework and Thematic Synthesis Plugin

- **Methodology:** Aligns with the call's emphasis on identifying recurring patterns and themes across diverse research designs.
- **Implementation:** The **Framework module** uses a *pre-specified theoretical framework* to guide extraction agents. The **Thematic module** identifies emergent themes through inductive coding and clustering, allowing for the synthesis of heterogeneous evidence bases into coherent categories.

4.2.3 Qualitative Comparative Analysis (QCA) Plugin

- **Methodology:** Supports the discovery of "causal configurations" and "contextual contingencies" specified in the RRE call.
- **Implementation:** Agents perform **case calibration**, transforming qualitative or quantitative data into fuzzy or crisp set memberships. The plugin identifies which combinations of conditions are necessary or sufficient for an outcome, enabling the system to reason about causal "paths" rather than isolated variables.

4.2.4 Meta-Ethnography and Critical Interpretive Synthesis (CIS) Plugin

- **Methodology:** Prioritizes theoretical innovation and concept translation across different methodological and disciplinary boundaries.
- **Implementation:** Agents perform **conceptual translation** across studies, identifying how concepts from one study can be reconciled or expanded by another to build a new overarching theory.

4.2.5 Meta-Analysis and Mixed-Method Synthesis Plugin

- **Methodology:** Integrates methodologically diverse evidence to inform systemic change.
- **Implementation:** Agents extract **effect sizes** and conduct risk-of-bias assessments for quantitative pooling. The mixed-method logic embeds text and

numeric data into **unified graph structures**, allowing agents to reason across diverse data types for causal discovery.

4.3 Human-in-the-Loop (HITL) and Lifelong Learning

The framework is governed by a formal **Master Control protocol** where human researchers act as the ultimate arbiters of synthesis logic through four predefined checkpoints:

1. **Screening Adjudication:** Reviewing AI-labeled samples to refine eligibility rules through few-shot calibration.
2. **Extraction Validation:** Spot-checking extracted CMOCs to ensure fidelity to the "ground truth" found in the literature.
3. **Contradiction Adjudication:** Reviewing a "Conflict Register" where same interventions lead to different outcomes, and providing interpretive explanations to resolve these contingencies.
4. **Theory Sign-off:** Approving the final synthesized programme theory and evidence map before artifact generation.

Every human intervention is recycled to update the system's **decision policies** (e.g., prompts and rules), enabling **lifelong (continual) and online learning**. This ensures that the framework accommodates evolving stakeholder requirements and matures its reasoning throughout the review lifecycle.

How does the realist synthesis plugin handle CMOC configurations?
How will you evaluate the framework's programme theory fidelity?
What analytical artifacts will be produced for the RRE repository?

too simple, lack sufficient details ,redo the above as the section of draft paper

2. Literature Review and Gap Analysis

2.1 The Paradigm Shift in AI-Augmented Evidence Synthesis Post-2023

The emergence of Large Language Models (LLMs) and multi-agent orchestration frameworks after 2023 has fundamentally reshaped the landscape of research synthesis. Prior to this "generative AI sensation," automation was largely confined to machine-learning-based citation screening and structured metadata extraction . In the post-2023 era, the focus has shifted toward **agentic workflows** that attempt to replicate the interpretive and reasoning-heavy tasks once reserved for human experts . Current research explores the capacity of LLMs to support high-recall abstract screening and the extraction of complex constructs from full-text corpora, often employing collaborative "two-reviewer" agent designs to resolve discordant outputs .

2.2 Critical Assessment of General AI-Assisted Literature Review Platforms

The market for AI-driven research tools has expanded rapidly with the launch of platforms such as **Elicit, Consensus, and Scite Assistant**. These platforms optimize

the "speed-to-summary" pipeline, allowing researchers to screen thousands of papers and generate report-like syntheses in a fraction of the traditional time . For instance, recent product updates in Elicit claim to support "Scaled Systematic Reviews" of up to 40,000 papers, integrating search, screen, and extraction into a unified parallelized task .

However, from the perspective of the **RRE Volume 50 methodological requirements**, these general literature review platforms exhibit several decisive disadvantages for synthesizing evidence in **complex educational systems**:

- **Methodological Opacity and Weak Warranting:** Unlike a formal systematic review, which must be **methodologically explicit and evaluable**, many general platforms utilize ad hoc prompting and "black-box" reasoning . This prevents researchers from tracing claims back to specific evidence spans with the auditability required for scholarly critique .
- **Suppression of Contradictions:** General platforms often optimize for narrative "consensus" or "smoothing over" contested findings . This fails to meet the needs of researchers examining **contextual contingencies**, where the primary goal is often to identify *why* results differ across settings .
- **Lack of Configurational Logic:** Current platforms typically lack the capacity to operationalize method-specific synthesis logics, such as the **Context–Mechanism–Outcome configurations (CMOCs)** foundational to realist synthesis . They primarily function as productivity tools rather than as **methodological infrastructures** capable of producing the structured, depositable artifacts required by journals like RRE .

2.3 Advances in Multi-Agent Orchestration and Knowledge Graphs

To address the "global" reasoning failures of traditional Retrieval-Augmented Generation (RAG), post-2023 research has prioritized **Graph-enhanced Retrieval (GraphRAG)** and **multi-agent orchestration** . Multi-agent systems (e.g., using **LangGraph**) enable specialized agents to collaborate on discrete tasks—such as screening, coding, and synthesis—while maintaining a shared "state" of the review protocol .

Furthermore, the construction of a **Literature Knowledge Graph (LKG)** allows agents to reason over connected claims and evidence paths rather than isolated text passages . This enables the discovery of **demi-regularities** and **causal configurations** by mapping how specific intervention resources trigger learner responses within varied educational contexts . By integrating **Microsoft Graph** as an enterprise-grade retrieval layer, these frameworks can ensure that every decision is logged and every PDF is managed within a traceable institutional repository .

2.4 The Methodological Gap: From Automation to Adjudicated Governance

The primary gap in the current evidence-synthesis ecosystem is the absence of an infrastructure that combines the **interpretive depth of human experts** with the **scalability of agentic AI**. While general platforms can accelerate search, they do not accommodate the **iterative refinement of programme theories** seen in human-led reviews like **Richmond et al. (2020)** .
The proposed **core-and-plugin framework** addresses this gap by separating method-agnostic "Review OS" functions (retrieval, provenance, registry, and audit) from method-specific synthesis logics . This framework uniquely prioritizes:

1. **Human-in-the-Loop (HITL) Adjudication:** Formalizing checkpoints for screening calibration, CMOC validation, and theory sign-off to prevent "automation bias" .
2. **Lifelong (Continual) Learning:** Incorporating human feedback at each stage to iteratively update **decision policies, prompts, and extraction rules** as new literature emerges and stakeholder requirements evolve .
3. **Auditability for Sustainable Change:** Producing repository-ready artifacts—including contradiction registers and provenance-linked CMOC tables—to ensure the synthesis is not a "one-time" report but a **living, auditable evidence map** suitable for driving systemic educational innovation .

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How do CMOCs help model configurations of causal mechanisms?
What analytical artifacts are required for the RRE repository?
Explain how the pipeline handles contradictions in educational research.

I ask for method section, what you did does not meet the instruction

3. Methodology: A Multi-Agent Framework for Generative AI-Augmented Research Synthesis

This section details the proposed multi-agent orchestration framework, designed to satisfy the **Review of Research in Education (RRE) 2026** requirements for methodological innovation and transparent warranting in complex educational systems. The framework is implemented as a stateful agentic pipeline using post-2023 orchestration technology (e.g., LangGraph), allowing it to replicate the interpretive depth of human expert reviews, specifically the realist synthesis conducted by **Richmond et al. (2020)**.

3.1 Architectural Principles: Core Review Infrastructure and Plugin Functions

The framework is partitioned into two distinct layers to ensure both methodological stability and flexibility.

- **The Core Review Infrastructure (Essential):** This "Review OS" provides the method-agnostic services required for any systematic review. It manages **corpus ingestion, stable study registries, deduplication, and the Human-in-the-Loop (HITL) control module**. It is powered by **Microsoft Graph as the enterprise retrieval layer**, ensuring that every retrieved document and decision is logged within a traceable institutional repository.
- **Method-Specific Plugins (Modular):** Plugins instantiate the distinct synthesis logic of a particular tradition. The **Realist Synthesis Plugin** used here operationalizes the **Initial Programme Theory (IPT)** and **Context–Mechanism–Outcome configurations (CMOCs)**. By swapping plugins, the same core infrastructure can support framework synthesis, thematic synthesis, QCA, or meta-ethnography.

3.2 Operational Agent Roles and Task Decomposition

The framework employs specialized agents that collaborate through a **Shared State Registry**, serving as the single source of truth for the entire pipeline.

Agent Role	Input	Expected Output	Agent Responsibility
Protocol & IPT Agent	Review Question +	IPT Schema + Extraction Rubrics	Encodes the "Initial Programme Theory" to guide search and

	Seeding Theory		extraction.
Search & Retrieval Agent	IPT Themes + Search Strings	Raw Publication Superset	Executes queries via **Microsoft Graph API** across project folders and databases.
Deduplication Agent	Raw Metadata	Unique Study Registry	Clusters records and assigns stable **StudyIDs** for auditability.
Title/Abstract Screener	Metadata + Eligibility Rules	include/exclude/uncertain Label	Performs first-stage classification with structured rationales.
Full-Text Eligibility Agent	PDFs + Protocol Rules	Final Inclusion Report	Audits full texts for methodological rigour and theoretical fit.
CMOC Extraction Agent	Full-Text Chunks + IPT Schema	LKG Triples (C, M-res, M-resp, O)	Converts passages into structured **Context, Mechanism, and Outcome** configurations.
Synthesis & Refinement Agent	LKG Subgraphs	Updated Programme Theory	Clusters CMOCs into demi-regularities and proposes theory revisions.
Contradiction Agent	CMOC Store	Conflict Register	Detects conflicting evidence (same resource, different outcomes) for human adjudication.
Reporting & Audit Agent	Shared State Logs	PRISMA Flow + Evidence Tables	Generates final artifacts for repository deposition.

3.3 Prompting and Enforcement of Methodological Logic

To ensure the Large Language Model (LLM) acts as a specialized research agent, the framework uses a multi-layered prompting strategy:

- **System Prompts:** Define the agent's professional identity (e.g., "Act as a specialist in RAMESES-compliant realist synthesis").
- **Role Prompts:** Constrain the agent's scope to specific tasks (e.g., "Extract only student-level mechanism responses").
- **Structure Enforcement:** Agents are guided by **schemas and templates** to output structured JSON or evidence tables rather than narrative prose, ensuring every claim is linked to a specific text span ID.

3.4 Human-in-the-Loop (HITL) Governance and Checkpoints

The framework is governed by a **Master Control protocol** where the system pauses for human adjudication at four critical checkpoints:

1. **Screening Adjudication:** Humans review a stratified sample of decisions to calibrate the agent's logic through **few-shot examples**.

2. **CMOC Validation:** Humans spot-check extracted configurations against the **"ground truth"** of the publications.
3. **Contradiction Resolution:** When the system identifies conflicting evidence, it stops to let the human provide an interpretive explanation.
4. **Theory Sign-off:** The researcher approves the final synthesized programme theory before the **Reporting Agent** generates repository artifacts.

3.5 Lifelong (Continual) and Online Learning Mechanism

This framework supports **lifelong learning** by recycling human feedback into the system's decision policies. When a researcher corrects an agent's output, the system does not just edit the text; it updates its **underlying prompts, rubrics, and extraction rules**. This allows the agents to "learn" the researcher's intent, enabling the system to accommodate evolving stakeholder requirements and improve its accuracy throughout the lifecycle of the "living" review.

3.6 Literature Knowledge Graph and GraphRAG Integration

To support the configuration-level reasoning required by Richmond et al. (2020), the framework constructs a **Literature Knowledge Graph (LKG)**. Unlike traditional RAG which retrieves isolated text passages, **Graph-enhanced Retrieval (GraphRAG)** allows agents to retrieve **subgraphs of connected claims**. This enables the framework to reason over complex relationships—how specific pedagogical resources trigger cognitive responses in varied student contexts—providing a transparent evidence path from theory claims back to verifiable source passages.

How do CMOCs help reveal causal mechanisms in complex systems?
Why use Microsoft Graph instead of just a vector store?
How does human feedback improve the AI's future screening efforts?

too simple, more details are needed

3. Methodology: A Multi-Agent Orchestration Framework

This section details the proposed agentic infrastructure, designed to replicate the interpretive and theory-driven workflow of the **Richmond et al. (2020)** realist review. The framework is architected as a stateful, human-governed system that separates method-agnostic "Review Infrastructure" from method-specific "Synthesis Plugins".

3.1 Architectural Philosophy: The Core-and-Plugin Model

The framework addresses the scalability and transparency limitations of traditional reviews by utilizing a **core-and-plugin architecture**.

- **Core Review Infrastructure (Method-Agnostic):** Provides foundational services such as enterprise-grade retrieval, deduplication, and a unified audit trail.
- **Method-Specific Plugins:** These modular extensions instantiate the unique logic of a synthesis tradition. This manuscript focuses on the **Realist Synthesis**

Plugin, which operationalizes the **Context–Mechanism–Outcome (CMO)** logic required to analyze complex educational systems.

3.2 Agent Roles, Inputs, and Prompting Strategy

The framework employs 10 specialized agents coordinated via a stateful orchestrator (e.g., LangGraph). Each agent operates under **role-specific system prompts** and **structured output schemas** to ensure methodological rigor.

Agent Role	Input	Expected Output	Prompt Strategy/Example
Search & Retrieval Agent	IPT Themes & Search Strings	Raw Publication Superset	**Role:** "Act as a librarian using **Microsoft Graph API** to harvest literature from project folders".
Deduplication Agent	Raw Metadata	Unique Study Registry	**Logic:** Clusters duplicate records and assigns stable **StudyIDs** for cross-agent tracking.
Title/Abstract Screener	Metadata + Eligibility Rules	include/exclude/uncertain Decision	**Schema:** "Output JSON with rationale and confidence score based on the following criteria...".
CMOC Extraction Agent	Full-Text Chunks + IPT Schema	LKG Triples (C, M-res, M-resp, O)	**Inference:** "Identify Context, Mechanism Resource, Mechanism Response, and Outcome. Link each to a specific text span".
Synthesis & Refinement	LKG Subgraphs	Updated Programme Theory	**Patterning:** "Identify demi-regularities across studies to refine the initial programme theory".
Contradiction Agent	CMOC Store	Conflict Register	**Critique:** "Flag instances where the same resource leads to different outcomes in different contexts".

3.3 The Role of Microsoft Graph and Literature Knowledge Graphs

The framework utilizes two distinct graph-based technologies to ground AI reasoning:

1. **Microsoft Graph (Enterprise Layer):** Functions as the **retrieval and provenance backbone**. It allows agents to securely access and version project artifacts, such as PDFs, screening logs, and human notes stored in Teams or SharePoint.
2. **Literature Knowledge Graph (LKG):** A method-facing representation built from the literature where nodes represent interventions, mechanisms, and outcomes. This enables **Graph-enhanced Retrieval (GraphRAG)**, allowing agents to reason over connected causal paths rather than isolated text snippets.

3.4 Master Control: Human-in-the-Loop (HITL) and Lifelong Learning

Crucial to the framework is the **Master Control protocol**, which ensures human decisions dominate the synthesis process. The system pauses for human adjudication at four predefined checkpoints:

- **Checkpoint 1: Screening Adjudication:** Humans review a stratified sample of the AI's screening decisions. Feedback is recycled as **few-shot examples** to refine the Screener Agent's policy before it re-runs the task.
- **Checkpoint 2: Extraction Validation:** Humans spot-check extracted CMOC tuples against the "ground truth" evidence spans.
- **Checkpoint 3: Contradiction Resolution:** When the Contradiction Agent detects conflicting evidence, it presents a **Conflict Register**. The human provides an interpretive explanation which is stored as a new rule for the Synthesis Agent.
- **Checkpoint 4: Theory Sign-off:** The lead researcher approves the final synthesized programme theory and evidence map before artifact generation.

Lifelong (Continual) Update Mechanism: Every human correction—whether a modified screening label or a refined CMOC—is used to iteratively update the system's **decision policies (prompts and rules)**. This ensures the framework adapts to evolving human requirements and creates an auditable record of how human expertise shaped the final synthesis.

3.5 Benchmark Evaluation Design (Richmond et al. 2020)

To verify the framework, we replicate the search space and synthesis products of the **Richmond et al. (2020)** review.

- **Search/Screening Evaluation:** The pipeline's included set is compared against Richmond's **28 studies**, computing recall (proportion of Richmond's studies identified) and precision.
- **Synthesis Fidelity:** We evaluate the correspondence of system-generated CMOC patterns and programme theory against the human-authored synthesis reported in the benchmark paper.
- **Artifact Transparency:** To satisfy the RRE call, the framework exports a full package of **analytical artifacts**, including logged queries, screening rationales, and LKG-derived configuration maps for deposition in the AERA-RRE repository.

How do these plugins specifically manage different synthesis logics?
Tell me more about the human-in-the-loop governance checkpoints.
How does GraphRAG improve reasoning compared to standard RAG?

Please only focus on "medical education" uploaded and read carefully the Richmond Medical Education literature review carefully, and give more detail on what essentially they give as well as a full explanation of what their procedure is. Give it in language that is spoken to the layman. Particularly say how they initiate the initial theory as a kind of hypothesis, and then they try to look for from the literature anything similar to this hypothesis, based on all relevant information, and then they

try to check if there is any contradiction, consolidate this contradiction, and then they come out with the theory that is verified by previous research, that is stated for what kind of population, what kind of intervention has some effect. And please give more detail, more detail. And eventually what their paper is about. They review 28 papers, and what they try to do, and what is their final result. Please provide more concrete details about their process. For each process, what they try to do, and what is the outcome from this particular process. And then the outcome from the previous process will become the input for the next process. And eventually, the final output identifies the effect on some people, the effect of certain pedagogy, how they interact with the people. It's the kind of thing like this. Please give more detail and understanding so that we can let the AI agent do this in a more specific way. And how this knowledge graph plays a role. How their work will turn into an automatic process powered by a large language model. Yeah, please read that paper carefully. How they start with the initial theory, how they identify keywords to search the literature, and what is their criteria to identify 28 studies. Such things like give more detail and eventually what the result is based on reviewing, summarizing, or synthesizing the outcomes from these 28 publications. What eventually the output is and what is the practical significance.

The research conducted by **Richmond and colleagues (2020)** in the field of medical education is a "realist review," which is a fancy way of saying they didn't just want to know *if* a teaching method works, but **why** it works for some students and not others, and in what specific situations.
1. The Starting Point: The "Initial Theory" (Hypothesis)
Before looking at a single research paper, the team started with a **hypothesis**, known in their field as an **Initial Programme Theory (IPT)**.

- **The Input:** They combined their own expertise as clinical teachers with a quick "scoping" search of existing ideas.
- **The Process:** They sketched out a rough map of how they *thought* medical students learn to make diagnoses. They focused on "Dual-Process Theory," which suggests we think in two ways: fast/intuitive (non-analytical) and slow/deliberate (analytical).
- **The Outcome:** A draft set of "if-then" statements (called CMOCs) that served as the roadmap for the entire project.

2. Searching the Literature: Finding the "Matches"
Once they had their hypothesis, they needed to see if the real world (the literature) agreed with them.

- **The Input:** They identified key themes from their IPT—like "pattern recognition," "deliberate practice," and "illness scripts".

- **The Process:** They searched four major databases for any study published since the year 2000.
- **The Outcome:** A mountain of papers that they had to narrow down. They used a **two-stage screening process**: first, they checked titles and abstracts to see if the papers were relevant to their theory; second, they read the full texts to ensure the research was high-quality and actually added something to the "theory building".

3. Sifting Through 28 Papers: Identifying Patterns and Contradictions
After their rigorous screening, they were left with **28 specific papers** that formed the core of their review.

- **The Input:** These 28 papers acted as the "evidence" for their trial.
- **The Process:** This is where it gets interesting. They looked for **Context-Mechanism-Outcome configurations (CMOCs)**. Think of this like a chemistry formula: **Context** (the type of student) + **Mechanism** (the teaching resource and the student's reaction) = **Outcome** (how well they diagnosed the patient).
- **Handling Contradictions:** They didn't ignore studies that disagreed. Instead, they looked for *why* they disagreed. For example, they found that an intervention that helped a "high-knowledge" student might actually confuse a "low-knowledge" student. This is called the "Expertise Reversal Effect".
- **The Outcome:** By re-analyzing early papers in light of new things they learned from later ones, they consolidated all these "puzzle pieces" into one final, verified theory.

4. The Final Result: What Actually Works?
The final output of their work identifies exactly how different teaching styles interact with different types of students.

- **The Population:** They found that the **student is the key** variable. They identified five specific types of student contexts, primarily based on their **pre-existing knowledge** and **self-confidence**.
- **The Pedagogy:** Effective teaching requires more than just giving facts; it requires "safe places" (like simulations) where students can make mistakes and receive immediate feedback.
- **The Interaction:**
    - **Novices ("Low-Knowledge"):** They need explicit explanations of an expert's thinking. Without it, they might feel panic or resentment.
    - **Developing Experts:** Once they have enough knowledge, they should be encouraged to use "non-analytical" (intuitive) reasoning, just like real doctors do.

5. Practical Significance: Why Does This Matter?
This paper proves that a **"one-size-fits-all" approach to medical education is a failure**. For teachers, the significance is that they must identify a student's baseline knowledge *before* choosing a teaching method. If you give a task that is too hard for a novice, you aren't just wasting time—you might actually be causing "negative learning outcomes" like faulty future reasoning.
6. Powering This with AI: The Role of the Knowledge Graph
To turn this human process into an automatic one powered by an AI agent, we use a **Literature Knowledge Graph (LKG)**.

- **How it works:** Instead of the AI just reading text, the LKG turns Richmond's 28 papers into a **web of relationships**.
- **The Nodes:** Each "node" in the graph is a specific student type, a teaching method, or a result.
- **The Edges:** The connections (edges) represent the "because of." For example: `[Low Knowledge Student]` --(is confused by)--> `[Observing Expert without Explanation]`.
- **The Agent Workflow:** The AI agents replicate Richmond's steps. A **Screening Agent** finds the papers; an **Extraction Agent** pulls out the C-M-O formulas; and a **Contradiction Agent** flags when the graph shows two different results for the same method.

By using **GraphRAG**, the AI doesn't just summarize a paper; it "reasons" through the graph to see how a specific context (like a student's low confidence) changes the whole outcome, exactly as the human researchers did.
How do AI agents handle inferred mechanisms from research texts?
Tell me more about the expertise reversal effect findings.
How is the knowledge graph used for detecting evidentiary contradictions?

How the work done by Richmond in their process can be better represented by a knowledge graph, particularly by using the Microsoft knowledge graph. The entity is a kind of big concept, and the relation could be a kind of thing related to the relation between entities. And how those knowledge graphs can be mapped or related to Richmond's human works. How their close alignment could be, how this knowledge graph will better match or align with their work? How can we verify by using their publication? The data available from their publication and use those data to verify our knowledge graph's correctness or the appropriateness of our knowledge graph. Because we will have a big concept as an entity, and if any Richmond's publication includes those entities and includes the type of relation that can guide us to identify the big concept, knowledge in the field or words as an entity in building the knowledge graph by using Microsoft graph through using some community related to

identification for identifying the entity. Just look at this, have a close look from this perspective.

The work performed by Richmond et al. (2020) is inherently relational, making it uniquely suited for representation through a **Literature Knowledge Graph (LKG)**. By utilizing **Microsoft Graph** as a retrieval backbone and a method-specific knowledge graph for reasoning, the framework can automate the identification of "big concepts" (entities) and their causal links (relations) with a high degree of alignment to the original human-led research.

1. Representing Richmond's "Big Concepts" as Entities

In a realist review, the core units of evidence are not isolated facts but **configurations**. A knowledge graph represents these configurations as nodes (entities) and edges (relations):

- **Entities (Big Concepts):** These correspond to Richmond's identified categories, such as **Context factors** (e.g., "Low Knowledge Student," "Self-Confidence"), **Mechanism Resources** (e.g., "Explicit Explanations," "Simulated Patients"), and **Outcomes** (e.g., "Diagnostic Accuracy," "Faulty Reasoning").
- **Relations (Causal Links):** Instead of flat text, the graph stores the causal paths Richmond theorized. For example, a relation might be `[Intervention] --triggers--> [Mechanism:Response]` or `[Context] --moderates--> [Outcome]`.
- **Community Identification:** Using **GraphRAG** logic, the system can identify "communities" of related entities. This mirrors Richmond's manual process of identifying **demi-regularities**—patterns that recur across multiple studies—by clustering related CMOC (Context-Mechanism-Outcome Configuration) subgraphs.

2. Alignment with Richmond's Human Workflow

The knowledge graph is the "reasoning substrate" that allows AI agents to replicate Richmond's iterative synthesis:

- **IPT to Graph Schema:** Richmond's **Initial Programme Theory (IPT)** serves as the initial schema for the graph, defining what types of entities and relations the agents should look for.
- **CMOCs as Subgraphs:** Every time Richmond identified a CMOC in one of the 28 papers, the AI creates a **provenance-linked subgraph**.
- **Contradiction and Sensitivity:** Richmond specifically looked for why the same intervention worked for some and not others. A knowledge graph makes this computable by flagging **graph motifs** where the same "Mechanism Resource" leads to divergent "Outcomes" based on different "Context" nodes.

3. The Role of Microsoft Graph as the Backbone
Microsoft Graph serves as the enterprise-grade **provenance and retrieval layer** rather than the synthesis logic itself.

- **Retrieval Traceability:** It provides the agents with governed access to the project's PDFs, screening logs, and human notes stored in institutional repositories (Teams/SharePoint).
- **Ground Truth Mapping:** Every entity and relation in the LKG is persistently linked via Microsoft Graph to a specific **Evidence Span** (a unique ID for a section of a document), ensuring that every claim in the synthesis is traceable back to the raw publication data.

4. Verification and Validation of Correctness
The framework uses Richmond's 2020 publication as the **"human reference standard"** to verify the appropriateness of the knowledge graph:

- **Study Set Overlap:** The framework is verified by its ability to correctly identify and include the same **28 studies** Richmond selected, using recall and precision metrics.
- **Synthesis Fidelity:** The "correctness" of the knowledge graph is evaluated by comparing the system-generated **Programme Theory** (the final graph state) against Richmond's narrative synthesis.
- **Data-Driven Verification:** If the LKG identifies the same "big concepts" (e.g., the "Expertise Reversal Effect") and the same "relations" (e.g., "Low knowledge leads to panic when observing experts without explanation") that Richmond reported, the graph is considered a valid representation of the research field.

By converting the **28 publications** into a structured, relational graph, the framework ensures that the AI does not just summarize text but performs the same **configurational reasoning** Richmond and his colleagues used to conclude their review.
How can AI agents handle the CMOC extraction process?
How does Microsoft Graph support the provenance of evidence?
What practical benefits does a knowledge graph offer researchers?

what is it meant by "demi-regularities"

In the context of a realist synthesis—such as the study conducted by Richmond et al. (2020)—**demi-regularities** refer to **recurrent patterns** of Context–Mechanism–Outcome configurations (CMOCs) that are identified by comparing multiple studies.

Because realist research acknowledges that causal relationships are not linear but are instead **generative and dependent on complexity**, these patterns are not viewed as universal laws. Instead, they represent observable trends in how specific intervention resources trigger particular cognitive or emotional responses (mechanisms) within varied student contexts to produce specific outcomes.
Key aspects of demi-regularities within this framework include:

- **Cross-Study Patterning:** The identification of these patterns is a core task of the **Synthesis Agent**, which performs comparisons across the literature to find where similar "configurations of causal processes" occur.
- **Theory Refinement:** Identifying demi-regularities allows researchers (or AI agents) to propose updates to an **Initial Programme Theory (IPT)**, eventually refining it into an overall programme theory that explains "what works, for whom, and in what circumstances".
- **Knowledge Graph Discovery:** In the proposed multi-agent framework, demi-regularities are discovered by querying a **Literature Knowledge Graph (LKG)** to identify subgraphs where specific contexts consistently moderate the relationship between an intervention resource and its outcome.
- **Addressing Contradictions:** Finding these patterns helps in resolving apparent contradictions in the data; for example, if an intervention resource leads to different outcomes, identifying a demi-regularity might reveal that the outcome is consistently moderated by a specific student context, such as their pre-existing level of knowledge.

Ultimately, the discovery of demi-regularities provides the **warranted evidence** needed to build a comprehensive theory of how educational interventions function in complex systems.
How do demi-regularities appear as patterns within a knowledge graph?
What are the specific student contexts Richmond identified in the final theory?
How does the Contradiction Agent resolve divergent outcomes for the same pedagogy?

By using a knowledge graph for different research studies, each knowledge graph entity will contain the ID from different research, right? And through entity identification for big concepts that are used in the Microsoft knowledge graph, we can find similar results from different research, or we can find contradictions among the different research through the sub-knowledge graph — that is, the sub-graph about the same nodes. Can we say it like this? The knowledge graph can particularly facilitate identifying similar trends across different research by using different IDs for the different nodes and relations that you mentioned above. With this ID embedded in the entities and relations, the same trends or similar trends found in different research can

be consolidated, and the contradictory parts of the research can also be easily identified. Is that correct?

Your description is correct and aligns closely with the proposed agentic framework's logic for managing complex research data. By representing literature as a **Literature Knowledge Graph (LKG)**, the framework transforms narrative text into a structured, relational reasoning substrate where every entity and relation is embedded with stable identifiers for provenance and cross-study comparison.

ID Embedding and Provenance

Each entity in the knowledge graph contains unique identifiers, such as **StudyIDs** and **EvidenceSpan** pointers, which link specific concepts (nodes) and causal claims (edges) back to their original research sources. This persistent metadata ensures that every claim in the synthesis is traceable to a verifiable section of a document, fulfilling the requirement for **methodological explicitness and auditability**.

Identifying Trends and "Demi-Regularities"

The framework uses unified entity identification for "big concepts"—such as specific student-level contexts (e.g., "low knowledge") or mechanism resources (e.g., "explicit explanations")—to find similar results across different studies. By querying **subgraphs** centered on these shared nodes, the **Synthesis Agent** identifies recurrent patterns known as **demi-regularities**. This process allows the system to consolidate similar trends found in disparate research into a coherent **programme theory**.

Surfacing and Resolving Contradictions

The use of subgraphs is particularly powerful for identifying contradictions. The **Contradiction Agent** performs **motif detection** to flag specific graph patterns where a similar intervention or mechanism resource leads to divergent outcomes in different studies.

- **Consolidation:** The graph identifies when research agrees on a causal path.
- **Conflict Detection:** When research disagrees, the graph surfaces the conflict as a relational discrepancy (e.g., same resource, different result).
- **Adjudication:** These contradictory subgraphs are presented to the human researcher in a **Conflict Register** for interpretive resolution, such as identifying a moderating contextual factor that explains the difference.

The Role of Microsoft Graph

In this workflow, **Microsoft Graph** serves as the **enterprise retrieval and provenance layer**. It is the API gateway that allows agents to securely fetch the raw research studies, screening logs, and team notes from institutional repositories like SharePoint or Teams to build the review-specific knowledge graph. While Microsoft Graph provides the "plumbing" to access the data, the LKG provides the "intellectual logic" to reason over it.

If you would like a detailed look at how these graph structures are used to generate evidence tables or theory diagrams, I can provide more information on the **Reporting & Artifact Agent**. Or, I could create a **report** summarizing the specific "big concepts" Richmond identified in their 28-paper review.
How will you present the guidelines for using the method?
What are the specific artifacts produced for the RRE repository?
How does the human-in-the-loop control handle contradiction adjudication?

what it is meant by "motif detection"

In the context of the proposed multi-agent framework, **motif detection** refers to a structural graph-analysis technique used by AI agents to identify specific, recurring patterns or configurations within a **Literature Knowledge Graph (LKG)**. Rather than searching for isolated keywords, the system uses motif detection to query the graph for complex relational structures that represent the logic of a realist review. Specifically, motif detection is utilized in the following ways:

- **Identifying Configurations:** It allows agents to recognize specific **Context–Mechanism–Outcome configurations (CMOCs)** by treating them as distinct "graph patterns". This mirrors the human process of identifying "demi-regularities"—recurrent patterns across multiple studies—by searching for subgraphs where specific context factors consistently influence outcomes.
- **Contradiction and Sensitivity Detection:** The **Contradiction Agent** uses motif detection to flag instances where the evidence is inconsistent. For example, it can identify a graph pattern where the **same intervention and resource** lead to **different outcomes** depending on the specific **student context** (such as low vs. high pre-existing knowledge).
- **Configuration-Level Reasoning:** By identifying these motifs, the framework can perform the "why, how, and for whom" reasoning central to the **Richmond et al. (2020)** study. For instance, it can detect the "Expertise Reversal Effect" by finding motifs where directive teaching resources improve outcomes for novices but frustrate or hinder experts.
- **Structured Adjudication:** When a specific motif representing a contradiction is detected, the system can pause the automated workflow and present that specific subgraph to a human researcher for resolution, ensuring that every interpretive decision is grounded in a traceable evidence path.

Essentially, motif detection transforms the narrative findings of 28 or more research papers into **computable graph motifs**, making the identification of causal trends and theoretical conflicts an auditable and systematic process.
How does GraphRAG specifically detect conceptual patterns and contradictions?
How can researchers verify the correctness of a knowledge graph?

What analytical artifacts does the framework export for the repository?

what is i meant by motif, is it special term in AI or CS

In the context of the framework described in the sources, a **motif** refers to a **specific, recurring structural pattern or sub-graph configuration** within a knowledge graph. **Motif detection** is the computational process used by AI agents to identify these patterns to perform high-level relational reasoning.
Within this multi-agent system, **motif detection** is primarily used to:

- **Identify configurations:** Agents treat **Context–Mechanism–Outcome configurations (CMOCs)** as distinct graph patterns, which allows them to recognize "demi-regularities" (recurrent trends) across multiple research papers.
- **Surface contradictions:** The system can flag discrepancies—such as the **same intervention and resource** resulting in **different outcomes** based on different contexts (e.g., student knowledge levels)—by identifying specific relational patterns in the graph.
- **Support configuration-level reasoning:** By querying for these motifs, agents can determine "why, how, and for whom" an intervention works, replicating the theory-driven logic used by human researchers like **Richmond et al.**.

Regarding whether it is a "special term" in **AI or Computer Science (CS)**, the sources frame it as a foundational technique within **graph-based AI reasoning** and **Graph-enhanced Retrieval-Augmented Generation (GraphRAG)**.
Information outside of the sources: In the broader fields of Computer Science and Network Science, the term **"network motif"** is a standard technical term referring to small sub-graphs that appear in a network significantly more frequently than would be expected by chance. These motifs are often considered the functional "building blocks" of complex networks, much like a recurring melody is a motif in music.
How do multi-agent systems handle implicit mechanisms in research?
Can you detail the human-in-the-loop checkpoints for theory refinement?
How does GraphRAG specifically improve contradiction detection in synthesis?

please directly answer to this: what is i meant by motif, is it special term in AI or CS

In the context of the proposed multi-agent framework, a **motif** refers to a **specific, recurring structural pattern or sub-graph configuration** within a knowledge graph.
Is "Motif" a Special Term in AI or Computer Science?
Yes, it is a technical term used in **AI and Computer Science**, particularly within the fields of **graph-based AI reasoning** and **Graph-enhanced Retrieval-Augmented**

Generation (GraphRAG). While traditional search methods look for isolated keywords, **motif detection** is a structural analysis technique that allows AI agents to identify complex relational patterns.

How Motifs are Used in the Framework

- **Identifying Configurations:** Agents treat **Context–Mechanism–Outcome configurations (CMOCs)** as distinct graph patterns or "motifs". This allows the system to recognize "demi-regularities" (recurrent trends) across different research papers by identifying shared sub-graph structures.
- **Detecting Contradictions:** The **Contradiction Agent** utilizes motif detection to flag relational discrepancies in the evidence. For example, it can automatically find a "motif" where a single intervention resource leads to two different outcomes based on different student contexts.
- **Enabling Relational Reasoning:** By querying for these motifs, the framework can perform the "why, how, and for whom" reasoning central to a realist review, such as identifying the "Expertise Reversal Effect" through specific graph patterns.

Information outside of the sources: In broader Computer Science and Network Science, the term **"network motif"** is a well-established standard for describing small sub-graphs that appear in a network more frequently than would be expected by chance, often serving as the functional building blocks of that network. You may want to independently verify this broader application in network theory.

How do multi-agent systems specifically improve contradiction handling in reviews?
What criteria verify the fidelity of AI-generated programme theories?
How does Microsoft Graph support the auditability of evidence spans?

in KG, identify subgraph similary is N-P prpobilem , is it hard?

In the context of the proposed multi-agent framework, the task of identifying **subgraph similarity** or specific relational patterns—referred to in the sources as **motif detection**—is indeed a computationally challenging problem, although the sources do not explicitly use the term "NP-hard."

The Technical Complexity (Expert Context)

Information outside of the sources: In computer science, identifying whether one graph is contained within another (subgraph isomorphism) or calculating the similarity between two subgraphs is a well-established **NP-hard problem**. This means that as the size and complexity of the Knowledge Graph (KG) grow, the time required to find exact matches increases exponentially, making it "hard" for traditional algorithms to solve efficiently for very large datasets.

How the Framework Addresses This "Hardness"

While the underlying problem is computationally intensive, the proposed framework manages this complexity through its specialized agentic architecture and **Graph-enhanced Retrieval-Augmented Generation (GraphRAG)**:

- **Motif Detection for Reasoning:** The framework uses **motif detection** to allow agents to recognize specific **Context–Mechanism–Outcome configurations (CMOCs)** as distinct graph patterns. This allows the **Synthesis Agent** and **Contradiction Agent** to focus on small, meaningful "motifs" rather than attempting to analyze the entire global graph at once.
- **GraphRAG and Subgraph Retrieval:** Instead of performing an exhaustive search of the entire literature database, the system uses **GraphRAG** to retrieve **subgraphs of related entities**. This effectively "prunes" the search space, allowing the LLM-based agents to reason over connected claims and evidence paths that are relevant to a specific query.
- **Identifying Similarity and Trends:** The framework facilitates identifying similar trends (demi-regularities) across different research studies by centering queries on shared nodes, such as a specific "Mechanism Resource" or "Context factor". By assigning stable identifiers (StudyIDs) to these nodes and relations, the agents can more easily consolidate similar trends and surface contradictory evidence within these specific sub-graph neighborhoods.

Practical Significance in the Framework
In a practical sense, while identifying subgraph similarity is "hard" in a general mathematical context, it is precisely what makes the **Literature Knowledge Graph (LKG)** superior to traditional text-based search. By turning the narrative findings of the 28 benchmark papers into a structured graph, the framework enables agents to detect **relational motifs**—such as the "Expertise Reversal Effect"—which would be nearly impossible to find consistently through naive keyword searches.
Note: The specific classification of this problem as "NP-hard" is a concept from broader computer science and is not explicitly discussed in the provided sources. You may wish to independently verify the computational theory of subgraph isomorphism for further technical depth.
How does GraphRAG specifically improve reasoning over flat text chunks?
Tell me more about the 'Expertise Reversal Effect' found by Richmond.
What specific human-in-the-loop checkpoints are built into the framework?

I want you to provide the additional info on the above, and argue the proposed method workable and implementable

The proposed multi-agent framework is a **stateful orchestration system** designed to automate the interpretive and labor-intensive processes of a systematic review while maintaining the methodological rigor of a human research team. The framework is

architected as a **core-and-plugin system**, separating a method-agnostic core review infrastructure from method-specific synthesis plugins.

I. Additional Framework Details: Agent Roles and Workflow

The system utilizes specialized agents coordinated through a shared state (the "single source of truth"), ensuring that all decisions are auditable and traceable.

- **Core Review Infrastructure (Always Present):**
  - **Protocol & Scope Agent:** Encodes researcher-defined questions, eligibility rules, and Initial Programme Theory (IPT) schemas.
  - **Search & Retrieval Agent:** Executes queries across databases and uses **Microsoft Graph** to harvest literature from enterprise folders (Teams/SharePoint) with full provenance logging.
  - **Deduplication & Record-Linking Agent:** Clusters duplicate records across sources and assigns stable StudyIDs.
  - **Screening Agents (Title/Abstract & Full Text):** Apply criteria to classify records as *include, exclude,* or *uncertain*, maintaining detailed rationales for every decision.
  - **Reporting & Artifact Agent:** Packages all decision logs, PRISMA counts, and evidence tables into a repository-ready format.
- **Realist Synthesis Plugin (Method-Specific):**
  - **CMOC Extraction Agent:** Deconstructs full-text passages into structured **Context–Mechanism–Outcome configurations (CMOCs)**, identifying both explicit and inferred mechanisms.
  - **Theory Synthesis & Refinement Agent:** Compares CMOCs across studies to identify **demi-regularities** (recurrent patterns) and proposes evidence-based updates to the programme theory.
  - **Contradiction & Sensitivity Agent:** Uses **motif detection** to flag instances where similar interventions lead to divergent outcomes, forcing explicit resolution by human researchers.

II. Argument for Workability and Implementability

The framework is highly workable and implementable using current post-2023 technologies for the following reasons:

1. **Technical Substrate:** The system is built on established orchestration frameworks like **LangChain and LangGraph**, which are specifically designed to manage stateful, multi-step agent workflows.
2. **Enterprise-Grade Retrieval:** Utilizing **Microsoft Graph** as the retrieval backbone ensures the system is not a "black box"; every retrieved PDF and search query is securely logged and versioned within institutional repositories, providing a durable audit trail.

3. **Enhanced Reasoning via Knowledge Graphs:** Unlike traditional "flat" text retrieval, the use of a **Literature Knowledge Graph (LKG)** and **GraphRAG** allows agents to reason over connected causal paths. This is particularly critical for identifying complex configurations (CMOCs) that naive summarization tools often overlook.
4. **Evidence of LLM Capability:** Recent empirical studies (post-2023) confirm that LLMs are already capable of high-accuracy screening and extraction when governed by "two-reviewer" collaborative workflows and cross-critique logic.
5. **Human Master Control:** The framework is implementable because it does not attempt "total automation". Instead, it uses **predefined checkpoints** (Screening Adjudication, CMOC Validation, Contradiction Adjudication, Theory Sign-off) to ensure human judgment remains the governing force over interpretive steps.
6. **Benchmarked Validity:** The framework's workability is verifiable against the **Richmond et al. (2020)** human reference standard. By replicating Richmond's set of **28 included studies** and their final programme theory, the system provides measurable recall and precision metrics to prove its effectiveness.
7. **Lifelong Update Capacity:** The system implements a **continual learning loop** where human feedback at each checkpoint is recycled to update the underlying prompts and decision rules. This ensures the framework accommodates evolving stakeholder requirements and adapts as the evidence base grows.

This architecture directly addresses the scalability and transparency gaps of traditional reviews, offering a transparent, auditable, and extensible infrastructure suitable for informing change in complex educational systems.

How does GraphRAG specifically detect conceptual patterns and lineages?
Could you explain the Expertise Reversal Effect in more detail?
What repository artifacts are exported by the Reporting Agent?

I means that motif is workable for subgraph similarity determination. please make it

The use of **motifs** within the Literature Knowledge Graph (LKG) provides a robust and implementable method for determining **subgraph similarity**, which is essential for identifying recurrent patterns across disparate research studies. In this framework, a motif is defined as a specific, recurring structural pattern or sub-graph configuration that represents the underlying logic of the synthesis method—most notably the **Context–Mechanism–Outcome configurations (CMOCs)** required for realist synthesis.

I. Workability of Motif Detection for Subgraph Similarity

Motif detection effectively transforms the task of identifying similar trends (demi-regularities) from a narrative search into a **computable graph-matching process**. The framework enables this through the following mechanisms:

- **Pattern Consolidation:** By embedding stable **StudyIDs** and **EvidenceSpan** identifiers within the entities (e.g., "low knowledge student") and relations (e.g., "triggers response"), the system can identify where similar configurations of evidence appear across different studies. When the same motif—such as a specific resource leading to a positive outcome for a certain population—appears in multiple study subgraphs, the system can consolidate these into a single "demi-regularity".
- **Contradiction Identification:** Conversely, if the system detects a motif where the same intervention resource leads to divergent outcomes, it identifies a relational discrepancy. This "conflict motif" signals to the human researcher that a contextual contingency (e.g., student confidence levels) may be moderating the outcome, mirroring the human reasoning performed in the **Richmond et al. (2020)** benchmark.
- **Contextual Reasoning:** The framework uses **Graph-enhanced Retrieval-Augmented Generation (GraphRAG)** to retrieve these subgraphs rather than isolated text chunks. This allows agents to reason over the **similarity of the configurations** themselves, which is more powerful than naive keyword-based similarity.

II. Implementation and Alignment with Generative AI
This method is implementable using post-2023 orchestration and graph technologies:

- **Agentic Orchestration:** The framework utilizes a **stateful orchestrator** (such as LangGraph) to manage specialized agents that populate the graph and perform motif queries.
- **GraphRAG Integration:** Modern GraphRAG techniques allow for the extraction of a Knowledge Graph from raw text and the use of its community structures to improve reasoning over narrative data.
- **Traceability:** Every edge in the motif remains linked to a specific evidence span via **Microsoft Graph**, fulfilling the **RRE Call's** requirement for methodologically explicit warranting and auditable analytical artifacts.

By representing the findings of the 28 benchmark papers as structured graph patterns, the framework ensures that **subgraph similarity** is not merely a statistical approximation but a theoretically grounded identification of the "why, how, and for whom" of educational interventions.
How does motif detection help identify contradictions in educational studies?
What specific student contexts did the Richmond review identify?
How can human-in-the-loop checkpoints improve the AI's extraction accuracy?

how to address NP problem for the above , is it NP problems or can be solved

In the context of the proposed multi-agent framework, identifying subgraph similarity and detecting motifs (such as **Context–Mechanism–Outcome configurations** or **CMOCs**) does involve significant computational complexity. While the general problem of **subgraph isomorphism** is recognized in computer science as **NP-hard**—meaning the time required to find exact matches can grow exponentially with graph size—the framework is designed with specific strategies to make this problem **solvable and implementable** for systematic reviews.

I should note that the classification of graph similarity as an "NP problem" is a concept from broader computer science and is not explicitly discussed within the provided sources; however, the framework's architecture directly addresses the practical challenges associated with this complexity.

The framework addresses the "hardness" of graph reasoning through the following mechanisms:

1. Targeted Motif Detection vs. Global Search

The framework does not attempt a brute-force search for any possible pattern across the entire global graph. Instead, it utilizes **motif detection** to look for small, predefined structural configurations that represent the logic of the synthesis.

- **Structured Schemas:** The agents are guided by a specific schema involving entities like **Context, Mechanism (Resource/Response), and Outcome**.
- **Query-Focused Summarization:** By searching for these specific "building blocks" (the CMOCs), the search space is significantly restricted compared to a general subgraph similarity task.

2. Graph-enhanced Retrieval-Augmented Generation (GraphRAG)

The framework utilizes **GraphRAG** to manage complexity by retrieving **subgraphs of related entities** rather than trying to process the entire Literature Knowledge Graph (LKG) at once.

- **Neighborhood Retrieval:** Agents reason over connected claims and evidence paths within specific local neighborhoods of the graph.
- **Community Summarization:** Drawing on the GraphRAG approach, the system builds "community structures" within the graph. This allows the agents to summarize and query high-level patterns across the corpus without needing to traverse every individual node-to-node path in a single operation.

3. Use of Graph Embeddings and Link Prediction

To address missing information and avoid the pitfalls of exact combinatorial matching, the framework employs **Graph embeddings and link prediction**.

- **Vector Representations:** Embeddings translate high-dimensional graph data into lower-dimensional vector representations. This allows the system to

identify "similar" results and "demi-regularities" across different studies based on semantic and structural proximity rather than purely identical subgraph matching.
- **Denoising:** These methods help to "denoise" the graph and complete missing edges, ensuring that the agents can identify patterns even when studies report evidence with varying levels of detail.

4. Human-in-the-Loop Adjudication (HITL)
The framework incorporates **Master Control checkpoints** where human researchers adjudicate the most complex or contradictory subgraphs.

- **Contradiction Registers:** When the **Contradiction Agent** flags a "conflict motif" (e.g., the same intervention leading to different outcomes), it pauses the automated pipeline.
- **Human Decision Rules:** Instead of the machine attempting to resolve every combinatorial ambiguity, the human researcher provides the final interpretive explanation for the "contextual contingency". This human intervention effectively "solves" the interpretive challenge that purely computational methods struggle with.

Summary of Workability
By focusing on **small relational motifs**, using **localized GraphRAG retrieval**, and leveraging **vector embeddings**, the framework transforms an theoretically "hard" NP problem into a **workable, auditable, and traceable** methodological process. This ensures that the system can scale to handle the 28 papers in the Richmond benchmark—or even tens of thousands of papers in an enterprise setting—while maintaining the interpretive depth required for high-stakes educational research.
How do motifs handle contradictions in research findings?
What is the computational challenge of identifying subgraph similarity?
How does Microsoft Graph support the provenance of evidence spans?

You should not mention the 28 papers. These 28 papers are just a particular verification database. For general application, there may be more than 28 papers — many more papers. Do not specifically say '28 papers.' Instead, just say that for practical application in systematic reviews, the collection of identified relevant papers is not too large and can be resolved by the proposed method using MOTIF (M-O-T-I-F) for those.

For practical application in systematic reviews, the collection of identified relevant papers typically reaches a volume that is manageable for high-depth, configuration-level analysis through the proposed multi-agent framework. Rather than performing

simple narrative summaries, the framework utilizes a **Literature Knowledge Graph (LKG)** to represent each study as a relational structure.
The core synthesis logic is executed through **MOTIF detection**, a computational method that identifies specific, recurring graph configurations representing **Context–Mechanism–Outcome patterns**. By embedding stable identifiers within each entity and relation, the agents use **MOTIF** patterns to consolidate similar trends across different research studies or to surface contradictions within specific sub-graph neighborhoods. This approach ensures that the final synthesis is not just a summary of a fixed dataset, but an **auditable, configuration-level discovery process** capable of handling the increasing volume and complexity of contemporary educational research.
Synthesis and Pattern Recognition via MOTIF Detection
The framework's ability to resolve complex evidence bases rests on its structural reasoning layer. Once the relevant literature has been identified and screened, the **Synthesis Agent** populates the LKG with extracted causal claims. This transformation from flat text to a relational graph allows the system to employ **MOTIF detection** for high-order analysis:

- **Consolidating Trends:** The system identifies "demi-regularities" by detecting **MOTIFs** that recur across the literature, even when studies originate from diverse contexts. These motifs represent the generative mechanisms that explain how specific educational interventions achieve their effects.
- **Surfacing Contradictions:** When the graph identifies a conflict—such as the same intervention resource leading to divergent outcomes—it flags this as a **contradiction motif**. This prevents the "smoothing over" of contested findings common in general-purpose AI summary tools.
- **Traceable Warranting:** Every node and edge within a detected **MOTIF** is persistently linked back to its original source via stable **StudyIDs** and **EvidenceSpan** pointers. This fulfills the requirement for methodologically explicit warranting by ensuring every claim in the final programme theory can be traced back to a specific passage in the primary literature.

By utilizing this **MOTIF-based approach**, the framework effectively addresses the challenges of scalability and transparency that often limit traditional manual reviews of large and heterogeneous literatures. The result is a **transparent and controllable pipeline** where the final output—identifying the effects of specific pedagogies on particular populations—is anchored in a rigorous, computable evidence base.
How does MOTIF detection resolve causal configurations in complex systems?
What makes MOTIF-based trend consolidation more auditable than traditional summaries?
How can MOTIF patterns identify contradictions in educational intervention research?

what is the method for identifying similar subgraph

In the multi-agent framework, the method for identifying similar subgraphs is primarily driven by **MOTIF detection** (also referred to as relational graph queries or pattern recognition) performed over a **Literature Knowledge Graph (LKG)**.
The identification process works through the following structured steps:

1. Representation of Evidence as Relational Patterns

Instead of processing flat text, the framework deconstructs studies into **Context–Mechanism–Outcome configurations (CMOCs)**, which are stored as first-class relational objects in the knowledge graph. Each CMOC is represented as a **subgraph motif** containing nodes for the population (context), the pedagogical intervention (mechanism resource), the learner's reaction (mechanism response), and the result (outcome).

2. Pattern Discovery via MOTIF Detection

The framework's **Synthesis Agent** uses **MOTIF detection** to identify recurrent configurations across the graph.

- **Consolidating Trends:** By querying for specific structural patterns that repeat across different StudyIDs, the system identifies **demi-regularities** (recurrent trends). For example, a motif might represent a specific type of student (e.g., "low knowledge") consistently experiencing "panic" when observing an expert without explanation.
- **Identifying Contradictions:** The system uses motif detection to flag **relational discrepancies**. A "conflict motif" is detected when the same intervention resource leads to divergent outcomes in different study subgraphs, signaling a contextual contingency that requires human adjudication.

3. Similarity Resolution through Embeddings and GraphRAG

To overcome the computational "hardness" of exact subgraph matching, the framework employs two advanced techniques:

- **Graph Embeddings:** The system uses **graph embeddings and link prediction** to denoise the data and complete missing edges, allowing it to identify "similar" configurations based on semantic and structural proximity rather than purely identical labels.
- **GraphRAG (Neighborhood Retrieval):** Instead of a global search, agents use **Graph-enhanced Retrieval-Augmented Generation (GraphRAG)** to retrieve local **subgraphs of related entities**. This enables the agents to reason over connected claims and evidence paths specific to a context factor or intervention type.

4. Traceability and Provenance

A critical component of this method is that every node and edge within an identified motif remains linked to a specific **EvidenceSpan** pointer. Through **Microsoft Graph**,

the system maintains a traceable "audit trail," ensuring that any identified trend or contradiction can be verified against the original text passages in the primary literature.

Note on Computational Complexity: Outside of the provided sources, the general task of identifying exact subgraph similarity is known in computer science as the **NP-hard** problem of subgraph isomorphism. The proposed framework makes this "workable" for systematic reviews by using structured templates (CMOC schemas) and localized GraphRAG retrieval to narrow the search space to theoretically relevant motifs.

How does the framework manage contradictory evidence across research studies?
What are the four human master control checkpoints in the system?
How does MOTIF detection help identify trends in educational research?

in terms of multi LLM agent frame, Provide a description of how the large language model will be used as an agent — through prompts — to perform the tasks of the required work or to automatically replace the human researchers in doing the work that Richmond and his colleagues did in that example publication.

In a multi-agent framework, Large Language Models (LLMs) act as the specialized "reasoning engines" within an orchestrated pipeline, performing the interpretive tasks once reserved for human researchers. By utilizing **role-specific system prompts** and **structured templates**, the LLM is transformed from a general chatbot into a series of coordinated agents—such as **Screening, Extraction, and Synthesis agents**—that replicate the realist review process conducted by **Richmond et al. (2020)**.

1. Task Decomposition: Automating the Richmond Workflow

The framework decomposes the labor-intensive steps of the Richmond review into discrete agentic responsibilities governed by researcher-defined prompts:

- **Protocol & IPT Construction:** Instead of human consensus-building, a **Protocol Agent** uses prompts to encode the review's scope and **Initial Programme Theory (IPT)** into a machine-readable schema. This schema directs all subsequent agent actions.
- **Two-Stage Screening:** Richmond's team manually screened titles, abstracts, and full texts. In this framework, a **Screening Agent** is prompted with eligibility rules to classify records as *include, exclude,* or *uncertain*. The LLM generates a **rationale** for every decision, ensuring the process is auditable rather than a "black box".
- **Evidence Extraction (The Core Realist Task):** The framework automates the extraction of **Context–Mechanism–Outcome configurations (CMOCs)**. An **Extraction Agent** identifies specific entities (e.g., "low-knowledge students") and relations (e.g., "how feedback triggers learning responses"). Every

extracted item is persistently linked to a specific text span to prevent hallucinations.
- **Synthesis and Refinement:** A **Synthesis Agent** replicates Richmond's cross-study comparison to identify **demi-regularities** (recurrent patterns). A **Contradiction Agent** uses structural motif detection to flag instances where the same intervention leads to different outcomes, forcing explicit resolution.

2. Prompting and Rule-Based Governance
The LLM is governed through a multi-layered prompting strategy that ensures it follows the specific "logic" of a systematic review:

- **System Prompts:** These define the agent's professional persona (e.g., "Act as an expert in RAMESES-compliant realist synthesis").
- **Domain-Adapted Templates:** Prompts include researcher-defined **schemas** that force the LLM to output structured data (e.g., JSON or tables) rather than free-form prose.
- **Few-Shot Calibration:** During the "usage phase," human researchers review a small sample of the model's decisions. These corrections are fed back into the agent's prompt as **few-shot examples**, allowing the model to "learn" the researcher's intent and refine its screening or extraction logic in real-time.

3. Replacing Human Labor with "Master Control" Checkpoints
While the framework uses LLMs to automate high-volume tasks, it does not remove human judgment entirely. Instead, it creates **Master Control checkpoints** where the agents must pause for human adjudication:

1. **Screening Calibration:** Humans review a sample of labels to refine the agent's screening policy.
2. **CMOC Validation:** Humans spot-check extracted configurations against the **"ground truth"** found in the original publications.
3. **Contradiction Adjudication:** The system stops when it detects conflicting evidence (e.g., the "Expertise Reversal Effect"), requiring a human to provide an interpretive explanation.
4. **Theory Sign-Off:** The lead researcher approves the final synthesized programme theory before the **Reporting Agent** generates artifacts for repository deposition.

4. Knowledge Graph and GraphRAG Integration
To achieve the "relational reasoning" Richmond's team used to link mechanisms to contexts, the framework utilizes **Graph-enhanced Retrieval-Augmented Generation (GraphRAG)**. Instead of the LLM reading isolated text snippets, it retrieves **subgraphs of connected claims**. This allows the agents to reason over

complex interactions—such as how a student's confidence levels (Context) moderate their response to a simulated patient (Mechanism)—ensuring the synthesis stays aligned with the **complex educational system** being studied.
How do prompts transform the LLM into a Synthesis agent?
What specifically is identified during the CMOC extraction process?
How does human feedback refine the agents' screening logic?

Please describe those in LangChain and GraphChain terms. Will GraphChain be used to build the multi-agent agentic framework, or will we just use LangChain? Please describe those terms and the interactive pipeline in terms of LangChain or GraphChain.

The proposed multi-agent framework is built on the **LangChain** ecosystem, utilizing **LangGraph** as the primary stateful orchestrator for managing the complex, iterative logic of a realist review. While "GraphChain" is not a term explicitly mentioned in the provided sources, the framework heavily incorporates **graph-based orchestration** and **GraphRAG** (Graph-enhanced Retrieval-Augmented Generation) to manage the relationship between data and agents.
The interactive pipeline and framework are described through the following terms and mechanisms:
1. LangChain: The Orchestration Framework
LangChain serves as the foundational framework for building the orchestrated workflows.

- **Multi-Agent Orchestration:** It allows specialized agents (Screening, Extraction, Synthesis, and Contradiction agents) to collaborate by maintaining a shared "state" of the review protocol.
- **Domain-Adapted Templates:** LangChain uses researcher-defined prompt templates to comprehensively encode rules and schema definitions for specific review tasks, ensuring every agent action is guided by the **Initial Programme Theory (IPT)**.

2. LangGraph: The Stateful State-Machine
The sources recommend **LangGraph** (a specialized extension within LangChain) as the "ideal" orchestrator for this project.

- **Stateful Orchestration:** Unlike a simple linear chain, LangGraph models the review as a **graph-based state machine**. This is essential for the **iterative loops** required in realist synthesis, such as moving from extraction back to re-searching when new mechanisms are inferred.

- **Shared State Registry:** All agents read from and write to a "single source of truth" (the **Shared State**), which includes the study registry, versioned programme theories, and audit logs.

3. The Interactive Pipeline (Human-in-the-Loop)
The interactive nature of the pipeline is managed through **Predefined Checkpoints** within the LangGraph workflow. At these points, the automated "chain" pauses to allow for human adjudication:

- **Screening Adjudication:** The Human reviews a sample of AI-labeled records to calibrate the model's behavior.
- **CMOC Validation:** Humans spot-check the extracted **Context–Mechanism–Outcome configurations (CMOCs)** to ensure they align with the original literature.
- **Contradiction Adjudication:** When the **Contradiction Agent** identifies a conflict (e.g., same intervention leading to different outcomes), the system flags it for human interpretation rather than "smoothing" the data.
- **Theory Sign-Off:** The final "Programme Theory" must be approved by the researcher before the pipeline generates the final artifacts.

4. Graph-Enhanced Retrieval (GraphRAG)
While LangGraph manages the *agent workflow*, **GraphRAG** manages the *evidence*.

- **Relational Retrieval:** Instead of retrieving isolated text passages, the agents use GraphRAG to retrieve **subgraphs of connected claims** from the **Literature Knowledge Graph (LKG)**.
- **Configuration Reasoning:** This allows the LLM agents to reason over complex interactions—such as how a specific "Context" node (e.g., student self-confidence) interacts with an "Intervention" node to change the "Outcome"—matching the high-level reasoning performed by human researchers like **Richmond et al. (2020)**.

In summary, you will use **LangChain** as the broad agent framework and **LangGraph** as the specific tool to build the stateful, interactive pipeline. This architecture ensures the review is not just automated, but is **auditable, traceable, and governed by human expertise** at every critical juncture.
How does the MOTIF method identify similar trends across research?
How are human feedback loops integrated into the LangChain framework?
How does the Microsoft Graph ensure auditability within the framework?

Based on the LangChain framework, please describe each agent function and how each of them will be prompted, and what the workflow is for simulating the

systematic review and the plug-in specifics. Could you clearly talk about how the Graph RAG is built on the build-up of your graph and how they perform the required tasks as we outlined above. Go back and look at what we talked about regarding the agents, the core framework, and the plug-in functions for each specific systematic review approach. Please combine them in detail.

4. Methodology: The Multi-Agent Orchestration Framework

The proposed framework is a stateful orchestration system built on **LangChain** and **LangGraph**, designed to automate the labor-intensive stages of a systematic review while preserving the interpretive depth required for complex educational systems. The architecture is partitioned into a method-agnostic **Core Review Infrastructure** and method-specific **Synthesis Plugins**, ensuring the system can support diverse methodological traditions as required by the **Review of Research in Education (RRE) 2026 call**.

4.1 The Core Review Infrastructure (Method-Agnostic)

The core infrastructure provides the foundational "Review Services" that manage data provenance, audit trails, and human-in-the-loop (HITL) governance.

- **Protocol & Scope Agent:** This agent encodes researcher-defined review questions, inclusion/exclusion criteria, and **Initial Programme Theory (IPT)** schemas into versioned prompt templates. **Prompting Strategy:** It is prompted to "Act as a methodology expert," using structured schemas (e.g., JSON) to define the boundaries of the review.
- **Search & Retrieval Agent (Microsoft Graph API):** Utilizing **Microsoft Graph** as the enterprise retrieval layer, this agent harvests literature from organizational repositories like Teams, SharePoint, and OneDrive. It logs every query and source to ensure absolute retrieval traceability and reproducibility.
- **Deduplication & Record-Linking Agent:** This agent clusters candidate records across multiple databases, assigns stable **StudyIDs**, and maintains the "Single Source of Truth" study registry.
- **Two-Stage Screening Agent:** A multi-stage classifier that applies eligibility rules at the title/abstract and full-text levels. **Prompting Strategy:** It is prompted with few-shot examples from human calibration rounds, requiring it to output an "include, exclude, or uncertain" label alongside a **structured rationale** and a confidence score.
- **Reporting & Audit Agent:** This agent automatically packages all decision logs, search queries, extraction tables, and human feedback records into repository-ready artifacts suitable for the **AERA-RRE repository**.

4.2 The Realist Synthesis Plugin (Worked Example)

While the core manages the workflow, plugins instantiate the specific synthesis logic. The **Realist Synthesis Plugin** operationalizes the logic of **Richmond et al. (2020)** to explain *why, how, and for whom* interventions work.

- **CMOC Extraction Agent:** This agent deconstructs full-text passages into **Context–Mechanism–Outcome configurations (CMOCs)**. **Prompting Strategy:** It is prompted to identify "Context" (student/teacher factors), "Mechanism Resource" (what the intervention provides), "Mechanism Response" (learner reaction), and "Outcome". It maps these into **relational triples** for the knowledge graph.
- **Theory Synthesis & Refinement Agent:** It compares CMOCs across studies to identify **demi-regularities** (recurrent patterns). It proposes evidence-based updates to the programme theory, linking every theoretical claim back to specific **evidence-span pointers** in the literature.
- **Contradiction & Sensitivity Agent:** This agent utilizes **MOTIF detection** to flag instances where the same intervention resource leads to divergent outcomes in different contexts. It forces explicit adjudication of "contextual contingencies," preventing the system from "smoothing over" contested findings.

4.3 Workflow Simulation: Replicating Richmond et al. (2020)

The workflow is a stateful graph where transitions are governed by completion of tasks and human approval.

1. **IPT Seeding:** The **Protocol Agent** formalizes the initial hypothesis (e.g., Dual-Process Theory in clinical reasoning).
2. **Retrieval & Registry:** The **Retriever Agent** harvests the 28 benchmark papers and associated metadata via **Microsoft Graph**.
3. **HITL Screening Calibration:** Researchers review a stratified sample of 10% of records; their corrections update the **Screening Agent's** prompt logic through few-shot learning.
4. **Configurational Extraction:** The **Extraction Agent** populates the **Literature Knowledge Graph (LKG)** with nodes and edges representing the evidence base.
5. **MOTIF-Based Synthesis:** The **Synthesis Agent** queries the graph for recurrent CMOC patterns (MOTIFs) to refine the overall programme theory.
6. **Contradiction Resolution:** The **Contradiction Agent** identifies the "Expertise Reversal Effect" (e.g., directive teaching helps novices but hinders experts) by flagging conflicting subgraphs for human interpretation.

4.4 Literature Knowledge Graph and GraphRAG

The **Literature Knowledge Graph (LKG)** serves as the reasoning substrate for the agents. It is constructed by extracting entities (Intervention, Population, Outcome, Context) and relations from full texts using **Named Entity Recognition (NER)** and **Relation Extraction (RE)**.
GraphRAG (Graph-enhanced Retrieval-Augmented Generation) improves upon traditional RAG by allowing agents to retrieve **subgraphs of connected claims** rather than isolated text passages. By building "community structures" within the LKG, GraphRAG enables agents to perform **global, configurational reasoning**—such as mapping how student self-confidence (Context) moderates the response to simulated patients (Mechanism)—which matches the interpretive depth of the original human review.

4.5 Human-in-the-Loop (HITL) and Lifelong Learning

The framework is governed by a formal protocol where human researchers act as arbiters at four predefined checkpoints: **Screening Adjudication, CMOC Validation, Contradiction Adjudication, and Theory Sign-off**. Every human correction is recycled to update the system's **decision policies (prompts and rules)**, enabling **lifelong (continual) and online updating**. This ensures the framework accommodates evolving stakeholder requirements and adapts as new studies are added to the "living" review.

How do LangChain agents simulate the Richmond realist review?
How does MOTIF detection resolve causal contradictions in the graph?
What specifically separates core infrastructure from method-specific synthesis plugins?

You talk about knowledge graph, but you did not talk about how they build it. That is a key part, how we identify entities. Who will play this role?

The construction of the **Literature Knowledge Graph (LKG)** is a multi-stage process handled by specialized agents within the framework, utilizing both automated extraction and human-in-the-loop validation.

1. Who Plays the Role? (Agent Responsibilities)

The identification and structuring of entities are distributed across three primary roles:

- **Extraction Agent (or CMOC Extraction Agent):** This agent is the primary "identifier." It processes full-text chunks to identify key educational constructs—such as interventions, populations, outcomes, and contexts—and the relationships between them. For the realist plugin, this agent specifically focuses on identifying **Contexts, Mechanism Resources, Mechanism Responses, and Outcomes**.
- **Normalization Agent:** After initial extraction, this agent performs **canonicalization**. It is responsible for de-duplicating and merging entities that

refer to the same concept but use different terminology, such as resolving "self-efficacy" and "confidence" into a single node to ensure graph consistency.

- **Human Researcher (HITL):** Humans act as the final authority. At a predefined **CMOC Validation checkpoint**, researchers spot-check the extracted entities and relations against the original evidence spans to ensure interpretive accuracy and correct any systematic misinterpretations by the agents.

2. How They Build It (The Extraction Pipeline)
The building of the graph follows a systematic pipeline:

- **NER and RE Processing:** The framework uses **Named Entity Recognition (NER)** and **Relation Extraction (RE)** to automatically process publications harvested from databases like ERIC or Scopus. These methods identify the "nodes" (entities like *intervention* or *outcome*) and the "edges" (relations like *intervention affects outcome*).
- **Structuring into Triples:** The extracted data is organized into **relational triples** (e.g., [Study]—contains—[CMOC] or [CMOC]—has_context—[ContextFactor]). Every entity and relation is persistently linked to an **EvidenceSpan pointer**, which maps back to the specific docID and text segment to ensure traceability.
- **Denoising and Completion:** Once the initial graph is formed, the system employs **graph embeddings and link prediction**. These computational techniques help "denoise" the graph and identify potentially missing edges, resulting in a more structured and complete knowledge base for subsequent reasoning.
- **GraphRAG Integration:** The resulting LKG serves as the reasoning substrate. Instead of retrieving isolated text, the **Synthesis Agent** uses **Graph-enhanced Retrieval-Augmented Generation (GraphRAG)** to retrieve **subgraphs** of these connected entities, allowing for configuration-level reasoning over the evidence.

By separating the **identification** (Extraction Agent), **refinement** (Normalization Agent), and **verification** (Human Researcher), the framework ensures the knowledge graph is both scalable and methodologically rigorous.
How will the Extraction Agent identify specific educational constructs?
Which entities and relations populate the Literature Knowledge Graph?
How do researchers validate the graph during CMOC checkpoints?